SUPPLEMENT TO: NEW EXPONENTS FOR THE GENERALIZED FERMAT EQUATION $x^2 + y^3 = z^n$

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This supplement gives the certificates and data tables used in the proofs of the paper. Its sections and tables are numbered S1, S2, ..., and the paper cites each table by this number at the step of the proof that uses it. Numbers without the prefix S refer to the paper.

For the four exclusions of Part II that rest on the largest enumerations, Propositions 7.7, 8.10, 9.6 and 9.10, the tables give data from which the conclusions can be checked with linear algebra over $\mathbb{F}_\ell$ and a finite enumeration. For each of them we print a basis of $\Sigma + \Theta$ or of the annihilator of Σ, written in the coordinates of the local classes; the local classes and levels that occur at each place; the classes of the allowed branch data for Proposition 7.7; and the data of the 64 configurations of Proposition 9.6(f). The coordinates are those of the programs [4], which also give the bases of the local groups in which the classes are written. The local classes and logarithms themselves are computed by the programs, with the error control of Appendix B.1. The tables also allow the configurations of region A at depth ≥ 2 for 27a1 and of region B at depth ≥ 2 for 288a1 to be recomputed, by a meet-in-the-middle search over the tuples of ramification points; the other disc tests are not reproduced here. For the exponent 13 (Section S9) the tables give the local images, the Selmer and S-unit dimensions, the class group certificates, and the counts of the 3-adic tests and enumerations, but not the local classes themselves.

The tables were generated from the stored outputs of the programs of [4].

S1. The exponent 11

The certificate of this section is used in the proof of Proposition 6.2.

	a	b_0	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8	b_9	b_{10}	i with $\Upsilon_1(a,\sigma^i c)=0$
Υ_1	1	0	0	4	3	3	1	3	1	0	4	4	
Υ_2	0	1	0	1	4	2	2	3	4	0	1	2	
Υ_3	0	0	1	1	2	0	2	0	4	0	1	4	
2	2	1	1	4	1	2	2	1	1	2	2	0	$3^\dagger$, 5
7	3	0	0	4	1	0	0	0	1	4	1	3	5^*
12	4	4	4	4	1	3	3	4	1	1	0	1	1^*, 2, 5, $8^\dagger$
17	0	3	3	4	1	1	1	3	1	3	4	4	$0^\dagger$, 5
22	1	2	2	4	1	4	4	2	1	0	3	2	5, 9, 10^*

TABLE S1. A certificate for the disc test of Proposition 6.2 for $E = 54a1$, with $K = \mathbb{Q}[x]/(G_{11}(x, 9261/8))$. At v_1, $k_{v_1} = \mathbb{F}_5$ and $x \equiv 0$. At v_2, $k_{v_2} = \mathbb{F}_5[x]/(\bar{g})$, $\bar{g} = x^{11}+3x^{10}+4x^9+x^8+3x^7+2x^6+4x^5+4x^4+2x^2+x+3$, and a unit class (via (3.1)) is printed as $(b_0, \dots, b_{10})$, standing for $\sum_k b_k\, x^{5^k}$, so σ acts by $(\sigma b)_k = b_{k-1}$. Rows $\Upsilon_1, \Upsilon_2, \Upsilon_3$: the annihilator of Σ in reduced echelon form. Row j_1: the class a of the point over j_1 at v_1, and the class c of the point Q over j_1 at v_2 with $x(Q) \equiv x \pmod{v_2}$. Last column: the i with $\Upsilon_1(a, \sigma^i c) = 0$. At these i, $\Upsilon_2 \neq 0$, except at *, where $\Upsilon_3 \neq 0$, and at †, the known point.

S2. THE EXPONENT 17

The tables of this section are used in the proofs of Propositions 7.5, 7.6 and 7.7.

	k with $a_k = 1$	c_5	c_5'
σ_1	19, 21, 25, 33, 61, 65, 67	0	0
σ_2	23, 29, 37, 41, 63, 65, 67	0	1
σ_3	27, 37, 45, 53, 59, 65, 67	0	0
σ_4	31, 41, 45, 65	0	0
σ_5	35, 49, 65, 67	0	0
σ_6	39, 45, 53, 63, 67	0	1
σ_7	43, 49, 53, 65	0	1
σ_8	47, 49, 53, 59, 63, 67	0	1
σ_9	55, 59, 61, 63	0	1
σ_{10}	57, 61, 63, 67	0	0
σ_{11}	–	1	1
$(\delta_{v_2}(R_2(1)), 0)$	39, 45, 61	1	0
$(\delta_{v_2}(R_2(2)), 0)$	65	1	0
$(\delta_{v_2}(R_2(3)), 0)$	39, 45, 61, 65	1	0
$(\delta_{v_2}(T), 0)$	–	1	0

TABLE S2. A basis of the image Σ of $\mathrm{Sel}^2(J/K_{54a1})$ at the places above 2, in reduced echelon form, and the four pairs of Proposition 7.5, in the coordinates of its proof. Each row lists the odd k with $a_k = 1$, and the coordinates c_5 and c_5' at 5 in L_{v_2} and L_{v_1}. All other coordinates are 0: at v_2 the coordinate at Π, the a_k with $k \leq 17$ and the a_k not listed, and at v_1 the coordinates at $1+i$, $2+i$ and $-1+2i$.

Place $v \mid 2$, (e_v, f_v)	$(4,1)$	$(2,1)$	$(4,2)$	$(2,2)$
K_{864c1}, P_0 over 1536				
$\dim \Theta_v = \dim J(K_v)[2]$	2	1	2	1
$[\mathcal{O}_v : \Lambda_v]$	2^2	2^2	2^3	2^2
ν at the points over 1536	3	4	3	4
largest separation $v_2(\xi - \xi')$	1	2	1	2
conjugates $g_v P_{0,v}$ allowed by the disc test	2	2	4	2
levels of the fixed terms ($g_v \neq 1$)	2	1	$1, 2, 1$	2
level of $\mathcal{L}_{g_v P_{0,v}}(\tau_0)$, $\tau_0 = 1, 3, 2, 6, 4, 12$	$2, 2, 3, 3, 4, 4$	$3, 3, 4, 4, 5, 5$	$2, 2, 3, 3, 4, 4$	$3, 3, 4, 4, 5, 5$
K_{864b1}, P_0 over -13824				
$\dim \Theta_v$	1	2	2	1
ν at the points over -13824	2	3	2	3
largest separation $v_2(\xi - \xi')$	1	2	1	2
points over -13824 with x generating K_v	2	2	8	2
levels of the fixed terms ($g_v \neq 1$)	1	1	$0, 0, 0, 0, 1, 1, 1$	2
level of $\mathcal{L}_{g_v P_{0,v}}(2^n u_0)$, $0 \leq n \leq 5$	$n + 1$	$n + 2$	$n + 1$	$n + 2$

TABLE S3. Local data for the known points of Propositions 7.6 and 7.7, with levels in Λ_v. The fixed terms are listed for the conjugates $g_v P_{0,v} \neq P_{0,v}$, and for 864b1 the level of the branch term is the same for all centres and all u_0.

n	centres at the places of type $(4,1)$	$(2,1)$	$(2,2)$	$(4,2)$	number	levels d_v	m	outcome
1–12	P_0	P_0	P_0	P_0	48	$n+1, n+2, n+2, n+1$	$n+1$	excluded
1–12	P_0	P_0	g	P_0	48	$n+1, n+2, 2, n+1$	2	excluded
1–12	P_0	g	$*$	P_0	96	$n+1, 1, *, n+1$	1	excluded
1–12	g	$*$	$*$	P_0	192	$1, *, *, n+1$	1	excluded
1–12	P_0	P_0	$*$	g'	96	$n+1, n+2, *, 1$	1	class w_0
1–12	P_0	g	$*$	g'	96	$n+1, 1, *, 1$	1	excluded
1–12	g	$*$	$*$	g'	192	$1, *, *, 1$	1	excluded
0	g	P_0	$*$	g_2''	8	$2, 2, 2, 1$	1	class w_0
0	the other allowed tuples				56		1 or 2	excluded
					832			

TABLE S4. The branch data allowed by the disc test in Proposition 7.7, with $n \leq 12$. At the first three places, where P_0 has two conjugates, a centre $g_v P_{0,v}$ is written P_0 if $g_v = 1$ and g otherwise. At the place of type $(4,2)$, $g'P_0$ is the conjugate other than $P_{0,v}$ with the class of $P_{0,v}$, and $g_1''P_0$, $g_2''P_0$ are the two conjugates for which the branch points around them with $n = 0$ have the class of $P_{0,v}$. For $n = 0$ the level d_v at this place is 2 for $g_1''P_0$ and 1 for $g_2''P_0$. An asterisk stands for either centre, or for the corresponding level (2 or $n+2$ at the place of type $(2,2)$, 1 or $n+2$ at the place of type $(2,1)$). The column "number" counts the branch data in the row, that is, the triples of a tuple (g_v), a class of u modulo 8 and a value of n. The levels are those of $\mathrm{Log}_v(Q - P_0)$ in Λ_v, at the places in the order of the columns, and "excluded" means that $\mathrm{cl}_m(S) \notin (\Sigma + \Theta)/\Theta$.

(i) A basis of $\Sigma + \Theta$ in reduced echelon form (blocks: v_1, v_2, v_3, v_4)

b_1	0100100001 00000000 0000000000 10000001111011110110
b_2	0011000101 00000000 0000000000 00000100010010101110
b_3	0000010101 00000000 0000000000 10000110011001010100
b_4	0000001001 00000000 0000101110 10000001111011110110
b_5	0000000011 00000000 0000000000 10000101101001011000
b_6	0000000000 10001001 0000000000 10000101011000101001
b_7	0000000000 00100000 0000000000 00000100010010101110
b_8	0000000000 00010001 0000000000 00000000000000000000
b_9	0000000000 00000011 0000000000 00000100010010101110
b_{10}	0000000000 00000000 0100000010 00000000000000000000
b_{11}	0000000000 00000000 0010111111 00000100100011011111
b_{12}	0000000000 00000000 0001100110 10000110101000100101
b_{13}	0000000000 00000000 0000000000 01000000000100001010
b_{14}	0000000000 00000000 0000000000 00100110010000111010
b_{15}	0000000000 00000000 0000000000 00010100000110010101
b_{16}	0000000000 00000000 0000000000 00001110000010000011

(ii) The level and class of $2^{-d_v}\,\mathrm{Log}_v(Q - P_0)$ at each place and centre

place	centre	$n = 0$	$n \geq 1$
v_1 (4,1)	P_0	$n+1$: 0000001010 (every n)	
	g_2P_0	$u_0 \in \{1,5\}$ 2: 0100111110; $u_0 \in \{3,7\}$ 2: 0100110100	1: 0000001010
v_2 (2,1)	P_0	$n+2$: 00010010 (every n)	
	g_2P_0	1: 00010010	1: 00010010
v_3 (2,2)	P_0	$n+2$: 0110111111 (every n)	
	g_2P_0	2: 0000101110	2: 0110010001
v_4 (4,2)	P_0	$n+1$: 00101111110100010001 (every n)	
	g_2P_0	0: 00000111100010100010	0: 00000111100010100010
	g_3P_0	0: 00000111100010100010	0: 00000111100010100010
	g_4P_0	0: 00000111100010100010	0: 00000111100010100010
	g_5P_0	0: 00000111100010100010	0: 00000111100010100010
	g_6P_0	$u_0 \in \{1,5\}$ 2: 00111010100001111100; $u_0 \in \{3,7\}$ 2: 00010101010101101101	1: 00101111110100010001
	g_7P_0	1: 00010100000110010101	1: 00111011110010000100
	g_8P_0	1: 00111011110010000100	1: 00010100000110010101

TABLE S5. Certificate for Proposition 7.7. The places $v_1, \ldots, v_4$ of $K = K_{864\mathrm{b}1}$ above 2 are listed with (e_v, f_v). Classes are written as strings of digits in the coordinates of the targets of the local maps δ_v used in [4], of lengths $10, 8, 10, 20$ at $v_1, \ldots, v_4$; Σ ($\dim 11$) and Θ ($\dim 6$) are given there in the same coordinates, and (i) is a basis of $\Sigma + \Theta$. In (ii), g_cP_0 is the conjugate of $P_{0,v}$ numbered c in [4] (at v_4, g_8, g_6, g_7 are g', g_1'', g_2'' of Table S4); the entry "d: c" gives the level $d = d_v$ and the class c of $2^{-d_v}\,\mathrm{Log}_v(Q - P_0)$ for $\tau = 2^n u$, $u \equiv u_0 \pmod 8$; the column $n \geq 1$ holds for every $n \geq 1$, for $n \geq 6$ by the argument of the proof. A branch datum allowed by the disc test (Table S4) chooses a centre at each place, u_0 and n; its level is $m = \min_v d_v$, and w is the tuple of the classes at the places with $d_v = m$, zero elsewhere. In (iii) each w is reduced by the basis (i): the remainder is nonzero, so $w \notin \Sigma + \Theta$, except for the class w_0 of the 104 branch data sent to the sieve.

(iii) The classes $w = \mathrm{cl}_m(S)$ of the 832 allowed branch data with $n \le 12$

m	number	w (blocks $v_1, \ldots, v_4$)	remainder modulo $\Sigma + \Theta$
1	104	0000000000 00000000 0000000000 0001010000110010101	0: this is w_0
1	104	0000000000 00010010 0000000000 0000000000000000000	nonzero at v_4, coordinate 6
1	104	0000000000 00010010 0000000000 0001010000110010101	nonzero at v_4, coordinate 6
1	104	0000001010 00000000 0000000000 0000000000000000000	nonzero at v_3, coordinate 5
1	104	0000001010 00000000 0000000000 0001010000110010101	nonzero at v_3, coordinate 5
1	104	0000001010 00010010 0000000000 0000000000000000000	nonzero at v_3, coordinate 5
1	104	0000001010 00010010 0000000000 0001010000110010101	nonzero at v_3, coordinate 5
2	44	0000000000 00000000 0110010001 0000000000000000000	nonzero at v_3, coordinate 5
2–13	48	0000001010 00000000 0000000000 0010111110100010001	nonzero at v_3, coordinate 5
2	4	0000001010 00000000 0110010001 0010111110100010001	nonzero at v_3, coordinate 9
2	2	0100110100 00010010 0000101110 0001010101010101101	nonzero at v_3, coordinate 5
2	2	0100110100 00010010 0110111111 0001010101010101101	nonzero at v_3, coordinate 9
2	2	0100111110 00010010 0000101110 0011101010000111100	nonzero at v_4, coordinate 8
2	2	0100111110 00010010 0110111111 0011101010000111100	nonzero at v_3, coordinate 5

TABLE S6. Certificate for Proposition 7.7, continued.

S3. The exponent 19

The tables of this section are used in the proofs of Propositions 8.3, 8.4 and 8.10.

Field	family, reference point	κ	a	$n_{\max}$	$\dim \Theta_v$	ν	separations	branch data
K_{864a1}	$-216+3^7\mathbb{Z}_3$, P_E	3^7	1	10	$0,1,1$	$\frac{9}{2},\frac{5}{2},\frac{5}{2}$	$\frac{5}{2},1,1$	264
K_{864b1}	$27+3^6\mathbb{Z}_3$, $\pm P_E$	3^6	2	10	$1,0,1$	$\frac{7}{2},\frac{3}{2},\frac{3}{2}$	$\frac{5}{2},1,1$	528
K_{54a1}	$-27+3^6\mathbb{Z}_3$, P_E	3^6	2	10	$0,1,1$	$\frac{9}{2},\frac{5}{2},\frac{5}{2}$	$\frac{3}{2},1,1$	264
	$378+3^6\mathbb{Z}_3$, P'_E	3^6	2	10	$0,1,1$	$\frac{9}{2},\frac{5}{2},\frac{5}{2}$	$\frac{3}{2},1,1$	264
	$-216+3^7\mathbb{Z}_3$, $R_0=P'_E-P_E$	3^7	1	10	$0,1,1$	$\frac{9}{2},\frac{5}{2},\frac{5}{2}$	$\frac{5}{2},1,1$	264

Table S7. The enumerations of Proposition 8.3, at the places with $(e,f)=(2,1),(6,1),(6,2)$. The values of ν and the separations are in v_3-units, the representatives are $\tau_0=3^n u_0$ with u_0 modulo 3^a and $n\le n_{\max}$ (Section 5.4), and the last column gives the number of branch data allowed by the disc test.

	b_{16}	b_{17}	b_{19}	b_{20}	b_{22}	b_{23}	b_{25}	b_{26}	b_{28}
a	−1	−1	−1	−1	−1	0	1	−1	−1
b_1	1	0	−1	−1	0	1	0	1	−1
b_2	1	0	0	−1	−1	−1	−1	0	−1
b_4	−1	1	0	−1	−1	0	−1	0	1
b_5	0	1	0	0	1	0	1	−1	1
b_7	−1	1	0	1	1	−1	1	0	0
b_8	−1	1	0	1	−1	1	−1	0	0
b_{10}	−1	1	0	1	1	0	−1	0	−1
b_{11}	0	0	0	1	−1	−1	0	1	−1
b_{13}	−1	1	0	0	0	0	−1	0	0
b_{14}	0	0	0	1	0	1	0	−1	0

Table S8. The image Σ of $\mathrm{Sel}(K_{96a1})$ at the places above 3 as a graph over $\mathbb{F}_3$: $\sigma\in\Sigma$ if and only if the row vector of the coordinates $b_{16},\dots,b_{28}$ of σ is the row vector $(a,b_1,\dots,b_{14})$ times this matrix (coordinates as in the proof of Proposition 8.4). Equivalently, the rows, completed by the unit vector of their label, form the reduced echelon basis of Σ.

(i) A basis of $\Sigma + \Theta$ in reduced echelon form (blocks: $v_1, \dots, v_6$)

b_1	0100 000 000002 000001 000022 001112	b_9	0000 000 000012 000000 001021 000211
b_2	0010 000 000002 000001 000022 001112	b_{10}	0000 000 000000 010100 000021 002212
b_3	0001 000 000000 000200 000000 000200	b_{11}	0000 000 000000 001100 000021 002212
b_4	0000 010 000002 000001 001012 002000	b_{12}	0000 000 000000 000012 002012 001210
b_5	0000 001 000002 000101 000002 001020	b_{13}	0000 000 000000 000000 012000 000000
b_6	0000 000 010001 000001 002002 002020	b_{14}	0000 000 000000 000000 000100 001210
b_7	0000 000 001001 000001 002002 002020	b_{15}	0000 000 000000 000000 000000 012000
b_8	0000 000 000100 000101 001000 002201		

(ii) The local outcomes for $P_0 = P_1$

n	place	outcomes "level: class"
$n = 0$	v_1	2: 0001, 2: 0002
	v_2	1: 012, 1: 021
	v_3	1: 000220, 1: 001211, 1: 011010, 1: 011220, 1: 012201, 1: 022110
	v_4	1: 000010, 1: 002102, 1: 010222, 1: 011020, 1: 022010, 1: 022120
	v_5	1: 001001, 1: 001111, 1: 011120, 1: 012001, 1: 020011, 1: 022210
	v_6	1: 000020, 1: 011020, 1: 011120, 1: 011210, 1: 022010, 1: 022210
$n = 1$	v_1	3: 0001, 3: 0002
	v_2	2: 012, 2: 021
	v_3	1: 020221, 1: 022100, 2: 011220, 2: 022110
	v_4	1: 011200, 1: 021012, 2: 011020, 2: 022010
	v_5	1: 012021, 1: 020221, 2: 011120, 2: 022210
	v_6	1: 022100, 2: 001211, 2: 011120, 2: 012201, 2: 022210
$n \geq 2$	v_1	$n+2$: 0001, $n+2$: 0002
	v_2	$n+1$: 012, $n+1$: 021
	v_3	1: 020221, 1: 022100, $n+1$: 011220, $n+1$: 022110
	v_4	1: 011200, 1: 021012, $n+1$: 011020, $n+1$: 022010
	v_5	1: 012021, 1: 020221, $n+1$: 011120, $n+1$: 022210
	v_6	1: 022100, 2: 020221, $n+1$: 011120, $n+1$: 022210

TABLE S9. Certificate for Proposition 8.10. The places $v_1, \dots, v_6$ of $K = K_{288a1}$ above 3 are those of the proof. Classes are strings of digits in the coordinates of the targets of the local maps δ_v used in [4], of lengths $4, 3, 6, 6, 6, 6$; Σ and Θ are given there in the same coordinates, and (i) is a basis of $\Sigma + \Theta$ ($\dim \Sigma = 10$, $\dim \Theta = 5$). In (ii) and (iii) the entries at v are the distinct pairs (d_v, class of $3^{-d_v} \operatorname{Log}_v S$) over the allowed pairs (ξ_v, ϱ_v), for $\tau = 3^n u$; they are the same for $u_0 = 1, 2, 4$. A level $n + k$ belongs to the centre $x(P_0)$. A branch datum chooses u_0, $n \leq 12$ and one entry at every place. Let m be its least level and w the tuple of the classes at the places of level m, zero elsewhere; reduction by (i) leaves a nonzero remainder in each of the 53184 branch data for P_1 and the 44928 for P_2, so $\mathrm{cl}_m(S) \notin (\Sigma + \Theta)/\Theta$. For P_2 and $n = 0$ no pair at v_1 is allowed.

(iii) The local outcomes for $P_0 = P_2$		
n	place	outcomes "level: class"
$n = 0$	v_1	none
	v_2	3: 012, 3: 021
	v_3	none
	v_4	none
	v_5	none
	v_6	none
$n = 1$	v_1	1: 0001, 1: 0002
	v_2	4: 012, 4: 021
	v_3	1: 000100, 1: 000200, 1: 022000, 1: 022200
	v_4	1: 000100, 1: 000200, 1: 002012, 1: 002112, 1: 010012, 1: 010212
	v_5	1: 000100, 1: 000200, 1: 001021, 1: 001221, 1: 011000, 1: 011100
	v_6	1: 000100, 1: 000200, 1: 010012, 1: 010212, 1: 022000, 1: 022200
$n = 2$	v_1	2: 0001, 2: 0002
	v_2	5: 012, 5: 021
	v_3	1: 022100, 2: 000100, 2: 000200, 2: 010012, 2: 010212
	v_4	1: 002212, 1: 010112, 2: 000100, 2: 000200
	v_5	1: 001121, 1: 011200, 2: 000100, 2: 000200
	v_6	1: 010112, 1: 022100, 2: 000100, 2: 000200
$n \geq 3$	v_1	n: 0001, n: 0002
	v_2	$n+3$: 012, $n+3$: 021
	v_3	1: 022100, 2: 010112, n: 000100, n: 000200
	v_4	1: 002212, 1: 010112, n: 000100, n: 000200
	v_5	1: 001121, 1: 011200, n: 000100, n: 000200
	v_6	1: 010112, 1: 022100, n: 000100, n: 000200

TABLE S10. Certificate for Proposition 8.10, continued.

S4. The exponent 23

The tables of this section are used in the proofs of Propositions 9.2, 9.6 and 9.10.

Υ_1	100000 000596 610A66 655344	Υ_4	000100 0002A4 A383A8 A96A97
Υ_2	010000 000288 311282 76882A	Υ_5	000010 000282 14A414 555572
Υ_3	001000 000690 46A584 932730	Υ_6	000001 000910 9863A0 530632
Υ_7	000000 10005A 341412 040009		
Υ_8	000000 010265 002556 880520		
Υ_9	000000 001312 A896A7 260732		

j_1	class c	zeros of $\Upsilon_1(\sigma^i c)$	j_1	class c	zeros of $\Upsilon_1(\sigma^i c)$
4	323425 225383 59A535 678884	9, 20	10	2506A7 A0A074 8337A6 394845	10
15	A47117 972351 658628 85A122	10	21	391898 795A96 620363 8409A7	4, 5, 11, 12, 17
26	660909 51A32A 716710 A31570	$0^\dagger$, 2, 16, 21^*	32	422A89 470908 418A20 2A7A59	3, 9, 16, 21
37	2846A0 1673A8 884803 113919	1^*, 14	43	56317A 15682A 205698 753000	8
48	9A8392 804376 9429A6 3A5267	4	54	6A4360 931741 0A2255 10A162	0, 15, 18
59	511084 451344 A00A99 587605	5, 13	65	735551 617663 99A912 666214	1, 13, 19
70	135876 0A9312 079081 769A53	7, 12, 18	76	876742 3A2585 78758A 012376	7, 14
81	859568 746390 137174 9403A1		87	907933 0884A7 574147 579428	2, 22
92	47225A 393369 2A5267 02274A	6, 11	98	A48024 863319 361804 A2558A	
103	096A41 A30337 36335A 204098	3^*, 17, 19^*	109	089215 549230 159471 481631	23
114	70A733 688305 421442 496436	8, 15^*, 22	120	11A406 224152 A46039 938793	

TABLE S11. The disc test of Proposition 9.2(b) for $E = 864a1$. Here $K = \mathbb{Q}[x]/(G_{23}(x, -216))$ has one place v above 11, and $k_v = \mathbb{F}_{11}[x]/(\bar{g})$ with $\bar{g} = x^{24}+10x^{23}+7x^{22}+9x^{21}+5x^{20}+7x^{19}+6x^{18}+8x^{17}+5x^{16}+7x^{15}+6x^{14}+7x^{13}+10x^{12}+6x^{11}+10x^{10}+5x^9+6x^8+6x^7+10x^5+5x^4+10x^3+6x^2+7x+4 \equiv G_{23}(x, -216) \pmod{11}$. A string $a_0a_1 \dots a_{23}$ of digits, with $\mathtt{A} = 10$, stands for $\sum_k a_k x^{11^k} \in k_v$, so the Frobenius $\sigma\colon z \mapsto z^{11}$ acts by $(\sigma a)_k = a_{k-1}$, indices modulo 24, and a row $\Upsilon = \upsilon_0 \dots \upsilon_{23}$ acts by $\Upsilon(a) = \sum_k \upsilon_k a_k$. The rows $\Upsilon_1, \dots, \Upsilon_9$ are the basis in reduced echelon form of the annihilator of Σ. The row j_1 gives the class $c = \delta_v(Q)$ of one point Q over the representative j_1, and the class tuples of the disc are the $\sigma^i c$, $0 \le i \le 23$. For $j_1 \equiv 4 \pmod{11}$, Q is the point with $x(Q) \equiv x$ modulo v; for $j_1 \equiv 10 \pmod{11}$, it is the point for which the coefficient vector $(r_0, \dots, r_{23})$ of $x(Q) \bmod v = \sum_k r_k x^k$, $0 \le r_k \le 10$, is lexicographically smallest. Another choice of Q replaces c by some $\sigma^k c$ and shifts the listed i by k. At the zeros i listed, $\Upsilon_2(\sigma^i c) \ne 0$, except at i^*, where $\Upsilon_3(\sigma^i c) \ne 0$, and at $i^\dagger$, where $\sigma^i c = \delta_v(P_E) \in \Sigma$.

(i) The annihilator of Σ in reduced echelon form (blocks: $v_1, \dots, v_{13}$)

Υ_1	1 0 00 00 00 00 60 06 50 69 58 0A A2	Υ_7	0 0 00 00 10 00 80 34 70 06 31 42 74
Υ_2	0 1 00 00 00 00 40 2A 21 75 A7 A3 81	Υ_8	0 0 00 00 01 00 90 79 53 A2 01 93 42
Υ_3	0 0 10 00 00 00 40 24 96 AA 35 61 97	Υ_9	0 0 00 00 00 10 A0 17 68 46 19 06 A3
Υ_4	0 0 01 00 00 00 10 41 49 32 10 21 70	Υ_{10}	0 0 00 00 00 01 40 88 10 27 42 08 36
Υ_5	0 0 00 10 00 00 70 88 6A 72 82 45 37	Υ_{11}	0 0 00 00 00 00 01 80 44 A8 89 05 33
Υ_6	0 0 00 01 00 00 60 72 A5 50 36 96 48		

(ii) The points ρ_i over 0, their classes, and the positions of ξ, $\iota_K\xi$

$\mathbb{Q}_{11}$	ρ_1: 9, ρ_2: 1
$\mathbb{Q}_{121}$	ρ_1: 90, ρ_2: 10, ρ_3: A9, ρ_4: 80, ρ_5: 33, ρ_6: 08, ρ_7: 80, ρ_8: 12
$(\xi_v, \iota_K\xi_v)$	v_1: (ρ_2,ρ_1), v_2: (ρ_1,ρ_2), v_3: (ρ_3,ρ_5), v_4: (ρ_8,ρ_6), v_5: (ρ_8,ρ_6), v_6: (ρ_4,ρ_7), v_7: (ρ_4,ρ_7), v_8: (ρ_6,ρ_8), v_9: (ρ_5,ρ_3), v_{10}: (ρ_5,ρ_3), v_{11}: (ρ_2,ρ_1), v_{12}: (ρ_1,ρ_2), v_{13}: (ρ_7,ρ_4)

TABLE S12. Certificate for Proposition 9.6. $K = K_{27a1}$ has the places v_1, v_2 above 11 with $K_v = \mathbb{Q}_{11}$ and $v_3, \dots, v_{13}$ with $K_v = \mathbb{Q}_{121}$; classes are in $k_v = \mathbb{F}_{11}$ or $\mathbb{F}_{11}[t]/(T)$ (one or two digits, A = 10), in the coordinates of [4], where Σ is also given. The ρ_i are numbered separately in $\mathbb{Q}_{11}$ and in $\mathbb{Q}_{121}$. The points over a rational j_0, their orbits (the $\mathbb{Q}_{11}$-irreducible factors of $G_{23}(x, j_0)$, numbered as in [4]) and their classes depend only on K_v, and are listed once for $\mathbb{Q}_{11}$ and once for $\mathbb{Q}_{121}$. (a): $S_0 = \{v_6, v_7, v_{13}\}$ is where ξ_v and $\iota_K\xi_v$ are distinct ρ_i with the same class. (b), region A at depth ≥ 2: by Lemma 4.5(d) the class tuple of a configuration is that of its ramification points, so the allowed configurations are the admissible choices, one point per place from the lists of (iii) with pairwise distinct orbits, whose tuple of ρ_i has its class tuple in Σ, that is, is killed by $\Upsilon_1, \dots, \Upsilon_{11}$. (d): the columns of (i) at the coordinates of S' are linearly independent. (e): for each pattern, the configurations of (iii) with the ramification points of the pattern have class tuples outside $\Sigma + k_{S'}$. (f): in each row $\vartheta^{\mathrm{f}} \notin \Sigma$ and $u\vartheta' + \vartheta^{\mathrm{f}} \notin \Sigma$ for $u \in \mathbb{F}_{11}^{\times}$, and $\vartheta' \notin \Sigma$ in the two rows with $S' = \emptyset$ (Table S14).

(iii) Region A, depth 1: the points near each ρ_i, as $\text{class}_{\text{orbit}}$	
$\bar{w} = 1$	$\mathbb{Q}_{11}$: ρ_1 0_2; ρ_2 1_1 $\mathbb{Q}_{121}$: ρ_1 79_{12} 92_{12}; ρ_2 10_{11} 10_{11}; ρ_3 03_7 38_9 55_{10}; ρ_4 $\mathtt{A1}_6$ 71_8 79_{13}; ρ_5 17_3 $\mathtt{A8}_4$ 95_5; ρ_6 54_3 23_4 46_5; ρ_7 $\mathtt{9A}_6$ $\mathtt{6A}_8$ 92_{13}; ρ_8 88_7 63_9 06_{10}
$\bar{w} = 2$	$\mathbb{Q}_{11}$: ρ_1 1_1; ρ_2 1_2 $\mathbb{Q}_{121}$: ρ_1 68_6 93_6; ρ_2 10_{11} 10_{11}; ρ_3 60_7 83_8 52_{10}; ρ_4 07_9 17_{12} 18_{13}; ρ_5 09_3 85_4 16_5; ρ_6 22_3 36_4 65_5; ρ_7 44_9 54_{12} 43_{13}; ρ_8 60_7 58_8 39_{10}
$\bar{w} = 3$	$\mathbb{Q}_{11}$: ρ_1 5_1; ρ_2 1_2 $\mathbb{Q}_{121}$: ρ_1 24_7 97_7; ρ_2 10_6 10_6; ρ_3 20_8 $\mathtt{8A}_{10}$ 96_{13}; ρ_4 $\mathtt{A4}_9$ $\mathtt{A9}_{11}$ 49_{12}; ρ_5 76_3 $\mathtt{2A}_4$ 04_5; ρ_6 15_3 31_4 77_5; ρ_7 67_9 12_{11} 62_{12}; ρ_8 20_8 91_{10} 35_{13}
$\bar{w} = 4$	$\mathbb{Q}_{11}$: ρ_1 8_1; ρ_2 1_2 $\mathbb{Q}_{121}$: ρ_1 $\mathtt{9A}_{13}$ $\mathtt{A1}_{13}$; ρ_2 10_6 10_6; ρ_3 84_8 41_{10} 70_{12}; ρ_4 75_7 31_9 35_{11}; ρ_5 41_3 56_4 02_5; ρ_6 $\mathtt{3A}_3$ $\mathtt{A5}_4$ 99_5; ρ_7 26_7 $\mathtt{2A}_9$ 96_{11}; ρ_8 47_8 $\mathtt{3A}_{10}$ 70_{12}
$\bar{w} = 5$	$\mathbb{Q}_{11}$: ρ_1 4_1; ρ_2 1_2 $\mathbb{Q}_{121}$: ρ_1 35_8 96_8; ρ_2 10_9 10_9; ρ_3 22_7 06_{10} 68_{13}; ρ_4 53_6 33_{11} 55_{12}; ρ_5 84_3 27_4 $\mathtt{A9}_5$; ρ_6 47_3 64_4 12_5; ρ_7 28_6 08_{11} 06_{12}; ρ_8 09_7 55_{10} 93_{13}
$\bar{w} = 6$	$\mathbb{Q}_{11}$: ρ_1 3_1; ρ_2 1_2 $\mathbb{Q}_{121}$: ρ_1 46_{10} 95_{10}; ρ_2 10_8 10_8; ρ_3 75_6 $\mathtt{3A}_7$ 91_9; ρ_4 08_{11} 28_{12} 06_{13}; ρ_5 92_3 $\mathtt{4A}_4$ 78_5; ρ_6 79_3 51_4 $\mathtt{A3}_5$; ρ_7 33_{11} 53_{12} 55_{13}; ρ_8 26_6 41_7 $\mathtt{8A}_9$
$\bar{w} = 7$	$\mathbb{Q}_{11}$: ρ_1 $\mathtt{A}_2$; ρ_2 1_1 $\mathbb{Q}_{121}$: ρ_1 $\mathtt{8A}_{11}$ 91_{11}; ρ_2 10_9 10_9; ρ_3 56_6 27_7 13_{13}; ρ_4 26_8 $\mathtt{2A}_{10}$ 96_{12}; ρ_5 25_3 64_4 10_5; ρ_6 86_3 27_4 10_5; ρ_7 75_8 31_{10} 35_{12}; ρ_8 $\mathtt{A5}_6$ 64_7 98_{13}
$\bar{w} = 8$	$\mathbb{Q}_{11}$: ρ_1 2_2; ρ_2 1_1 $\mathbb{Q}_{121}$: ρ_1 57_{13} 94_{13}; ρ_2 10_6 10_6; ρ_3 18_9 77_{10} 01_{12}; ρ_4 62_7 12_8 67_{11}; ρ_5 $\mathtt{A0}_3$ 62_4 47_5; ρ_6 $\mathtt{A0}_3$ 49_4 84_5; ρ_7 49_7 $\mathtt{A9}_8$ $\mathtt{A4}_{11}$; ρ_8 43_9 04_{10} $\mathtt{AA}_{12}$
$\bar{w} = 9$	$\mathbb{Q}_{11}$: ρ_1 6_2; ρ_2 1_1 $\mathbb{Q}_{121}$: ρ_1 13_{11} 98_{11}; ρ_2 10_7 10_7; ρ_3 14_6 45_{10} 37_{13}; ρ_4 44_8 43_9 54_{12}; ρ_5 68_3 91_4 50_5; ρ_6 93_3 $\mathtt{8A}_4$ 50_5; ρ_7 07_8 18_9 17_{12}; ρ_8 87_6 $\mathtt{A6}_{10}$ 74_{13}
$\bar{w} = 10$	$\mathbb{Q}_{11}$: ρ_1 7_1; ρ_2 1_2 $\mathbb{Q}_{121}$: ρ_1 02_7 99_7; ρ_2 10_{10} 10_{10}; ρ_3 42_9 $\mathtt{6A}_{11}$ 94_{13}; ρ_4 $\mathtt{9A}_6$ 92_8 $\mathtt{6A}_{12}$; ρ_5 $\mathtt{5A}_3$ 81_4 79_5; ρ_6 61_3 $\mathtt{7A}_4$ 92_5; ρ_7 $\mathtt{A1}_6$ 79_8 71_{12}; ρ_8 29_9 71_{11} 57_{13}

TABLE S13. Certificate for Proposition 9.6, continued.

β, S'	ϑ'	ϑ^{f}
P_0 $\emptyset$	0 0 49 40 72 24 07 35 25 28 04 68 70	–
P_0 $\{3\}$	0 0 62 40 72 24 07 35 25 28 04 68 70	0 0 73 00 00 00 00 00 00 00 00 00 00
P_0 $\{11\}$	0 0 49 40 72 24 07 35 25 28 77 68 70	0 0 00 00 00 00 00 00 00 00 73 00 00
P_0 $\{1,2\}$	0 0 49 40 72 24 07 35 25 28 04 68 70	2 9 00 00 00 00 00 00 00 00 00 00 00
P_0 $\{3,11\}$	0 0 62 40 72 24 07 35 25 28 77 68 70	0 0 73 00 00 00 00 00 00 00 73 00 00
P_0 $\{5,6\}$	0 0 49 40 24 72 07 35 25 28 04 68 70	0 0 00 00 38 83 00 00 00 00 00 00 00
P_0 $\{9,10\}$	0 0 49 40 72 24 07 35 28 25 04 68 70	0 0 00 00 00 00 00 00 72 49 00 00 00
P_0 $\{1,2,3\}$	0 0 62 40 72 24 07 35 25 28 04 68 70	2 9 73 00 00 00 00 00 00 00 00 00 00
P_0 $\{1,2,11\}$	0 0 49 40 72 24 07 35 25 28 77 68 70	2 9 00 00 00 00 00 00 00 00 73 00 00
P_0 $\{3,5,6\}$	0 0 62 40 24 72 07 35 25 28 04 68 70	0 0 73 00 38 83 00 00 00 00 00 00 00
P_0 $\{3,9,10\}$	0 0 62 40 72 24 07 35 28 25 04 68 70	0 0 73 00 00 00 00 00 72 49 00 00 00
P_0 $\{5,6,11\}$	0 0 49 40 24 72 07 35 25 28 77 68 70	0 0 00 00 38 83 00 00 00 00 73 00 00
P_0 $\{9,10,11\}$	0 0 49 40 72 24 07 35 28 25 77 68 70	0 0 00 00 00 00 00 00 00 72 49 73 00 00
P_0 $\{1,2,3,11\}$	0 0 62 40 72 24 07 35 25 28 77 68 70	2 9 73 00 00 00 00 00 00 00 73 00 00
P_0 $\{1,2,5,6\}$	0 0 49 40 24 72 07 35 25 28 04 68 70	2 9 00 00 38 83 00 00 00 00 00 00 00
P_0 $\{1,2,9,10\}$	0 0 49 40 72 24 07 35 28 25 04 68 70	2 9 00 00 00 00 00 00 72 49 00 00 00
P_0 $\{3,5,6,11\}$	0 0 62 40 24 72 07 35 25 28 77 68 70	0 0 73 00 38 83 00 00 00 00 73 00 00
P_0 $\{3,9,10,11\}$	0 0 62 40 72 24 07 35 28 25 77 68 70	0 0 73 00 00 00 00 00 72 49 73 00 00
P_0 $\{5,6,9,10\}$	0 0 49 40 24 72 07 35 28 25 04 68 70	0 0 00 00 38 83 00 00 72 49 00 00 00
P_0 $\{1,2,3,5,6\}$	0 0 62 40 24 72 07 35 25 28 04 68 70	2 9 73 00 38 83 00 00 00 00 00 00 00
P_0 $\{1,2,3,9,10\}$	0 0 62 40 72 24 07 35 28 25 04 68 70	2 9 73 00 00 00 00 00 72 49 00 00 00
P_0 $\{1,2,5,6,11\}$	0 0 49 40 24 72 07 35 25 28 77 68 70	2 9 00 00 38 83 00 00 00 00 73 00 00
P_0 $\{1,2,9,10,11\}$	0 0 49 40 72 24 07 35 28 25 77 68 70	2 9 00 00 00 00 00 00 00 72 49 73 00 00
P_0 $\{3,5,6,9,10\}$	0 0 62 40 24 72 07 35 28 25 04 68 70	0 0 73 00 38 83 00 00 72 49 00 00 00
P_0 $\{5,6,9,10,11\}$	0 0 49 40 24 72 07 35 28 25 77 68 70	0 0 00 00 38 83 00 00 72 49 73 00 00
P_0 $\{1,2,3,5,6,11\}$	0 0 62 40 24 72 07 35 25 28 77 68 70	2 9 73 00 38 83 00 00 00 00 73 00 00
P_0 $\{1,2,3,9,10,11\}$	0 0 62 40 72 24 07 35 28 25 77 68 70	2 9 73 00 00 00 00 00 72 49 73 00 00
P_0 $\{1,2,5,6,9,10\}$	0 0 49 40 24 72 07 35 28 25 04 68 70	2 9 00 00 38 83 00 00 72 49 00 00 00
P_0 $\{3,5,6,9,10,11\}$	0 0 62 40 24 72 07 35 28 25 77 68 70	0 0 73 00 38 83 00 00 72 49 73 00 00
P_0 $\{1,2,3,5,6,9,10\}$	0 0 62 40 24 72 07 35 28 25 04 68 70	2 9 73 00 38 83 00 00 72 49 00 00 00
P_0 $\{1,2,5,6,9,10,11\}$	0 0 49 40 24 72 07 35 28 25 77 68 70	2 9 00 00 38 83 00 00 72 49 73 00 00
P_0 $\{1,2,3,5,6,9,10,11\}$	0 0 62 40 24 72 07 35 28 25 77 68 70	2 9 73 00 38 83 00 00 72 49 73 00 00

TABLE S14. Certificate for Proposition 9.6(f): the 64 configurations of region B at depth 1 with $\bar{w} = 5$, with the known point β whose Selmer class they carry, the set S' (indices of the places), and the tuples ϑ', ϑ^{f} in the blocks $v_1, \dots, v_{13}$ and the coordinates of Table S12. In each row $\vartheta^{\mathrm{f}} \notin \Sigma$ and $u\vartheta' + \vartheta^{\mathrm{f}} \notin \Sigma$ for $u \in \mathbb{F}_{11}^{\times}$, and $\vartheta' \notin \Sigma$ in the two rows with $S' = \emptyset$, as $\Upsilon_1, \dots, \Upsilon_{11}$ show.

β, S'	ϑ'	ϑ^{f}
$\iota_K P_0$ $\emptyset$	0 0 28 86 35 44 97 72 40 49 68 04 70	–
$\iota_K P_0$ $\{10\}$	0 0 28 86 35 44 97 72 40 62 68 04 70	0 0 00 00 00 00 00 00 00 00 73 00 00 00
$\iota_K P_0$ $\{12\}$	0 0 28 86 35 44 97 72 40 49 68 77 70	0 0 00 00 00 00 00 00 00 00 00 00 73 00
$\iota_K P_0$ $\{1, 2\}$	0 0 28 86 35 44 97 72 40 49 68 04 70	9 2 00 00 00 00 00 00 00 00 00 00 00 00
$\iota_K P_0$ $\{3, 4\}$	0 0 25 53 35 44 97 72 40 49 68 04 70	0 0 49 59 00 00 00 00 00 00 00 00 00 00
$\iota_K P_0$ $\{7, 8\}$	0 0 28 86 35 44 59 24 40 49 68 04 70	0 0 00 00 00 00 58 38 00 00 00 00 00 00
$\iota_K P_0$ $\{10, 12\}$	0 0 28 86 35 44 97 72 40 62 68 77 70	0 0 00 00 00 00 00 00 00 00 73 00 73 00
$\iota_K P_0$ $\{1, 2, 10\}$	0 0 28 86 35 44 97 72 40 62 68 04 70	9 2 00 00 00 00 00 00 00 00 73 00 00 00
$\iota_K P_0$ $\{1, 2, 12\}$	0 0 28 86 35 44 97 72 40 49 68 77 70	9 2 00 00 00 00 00 00 00 00 00 00 73 00
$\iota_K P_0$ $\{3, 4, 10\}$	0 0 25 53 35 44 97 72 40 62 68 04 70	0 0 49 59 00 00 00 00 00 00 73 00 00 00
$\iota_K P_0$ $\{3, 4, 12\}$	0 0 25 53 35 44 97 72 40 49 68 77 70	0 0 49 59 00 00 00 00 00 00 00 00 73 00
$\iota_K P_0$ $\{7, 8, 10\}$	0 0 28 86 35 44 59 24 40 62 68 04 70	0 0 00 00 00 00 58 38 00 73 00 00 00 00
$\iota_K P_0$ $\{7, 8, 12\}$	0 0 28 86 35 44 59 24 40 49 68 77 70	0 0 00 00 00 00 58 38 00 00 00 73 00 00
$\iota_K P_0$ $\{1, 2, 3, 4\}$	0 0 25 53 35 44 97 72 40 49 68 04 70	9 2 49 59 00 00 00 00 00 00 00 00 00 00
$\iota_K P_0$ $\{1, 2, 7, 8\}$	0 0 28 86 35 44 59 24 40 49 68 04 70	9 2 00 00 00 00 58 38 00 00 00 00 00 00
$\iota_K P_0$ $\{1, 2, 10, 12\}$	0 0 28 86 35 44 97 72 40 62 68 77 70	9 2 00 00 00 00 00 00 00 73 00 73 00 00
$\iota_K P_0$ $\{3, 4, 7, 8\}$	0 0 25 53 35 44 59 24 40 49 68 04 70	0 0 49 59 00 00 58 38 00 00 00 00 00 00
$\iota_K P_0$ $\{3, 4, 10, 12\}$	0 0 25 53 35 44 97 72 40 62 68 77 70	0 0 49 59 00 00 00 00 00 73 00 73 00 00
$\iota_K P_0$ $\{7, 8, 10, 12\}$	0 0 28 86 35 44 59 24 40 62 68 77 70	0 0 00 00 00 00 58 38 00 73 00 73 00 00
$\iota_K P_0$ $\{1, 2, 3, 4, 10\}$	0 0 25 53 35 44 97 72 40 62 68 04 70	9 2 49 59 00 00 00 00 00 73 00 00 00 00
$\iota_K P_0$ $\{1, 2, 3, 4, 12\}$	0 0 25 53 35 44 97 72 40 49 68 77 70	9 2 49 59 00 00 00 00 00 00 00 00 73 00
$\iota_K P_0$ $\{1, 2, 7, 8, 10\}$	0 0 28 86 35 44 59 24 40 62 68 04 70	9 2 00 00 00 00 58 38 00 73 00 00 00 00
$\iota_K P_0$ $\{1, 2, 7, 8, 12\}$	0 0 28 86 35 44 59 24 40 49 68 77 70	9 2 00 00 00 00 58 38 00 00 00 73 00 00
$\iota_K P_0$ $\{3, 4, 7, 8, 10\}$	0 0 25 53 35 44 59 24 40 62 68 04 70	0 0 49 59 00 00 58 38 00 73 00 00 00 00
$\iota_K P_0$ $\{3, 4, 7, 8, 12\}$	0 0 25 53 35 44 59 24 40 49 68 77 70	0 0 49 59 00 00 58 38 00 00 00 73 00 00
$\iota_K P_0$ $\{1, 2, 3, 4, 7, 8\}$	0 0 25 53 35 44 59 24 40 49 68 04 70	9 2 49 59 00 00 58 38 00 00 00 00 00 00
$\iota_K P_0$ $\{1, 2, 3, 4, 10, 12\}$	0 0 25 53 35 44 97 72 40 62 68 77 70	9 2 49 59 00 00 00 00 00 73 00 73 00 00
$\iota_K P_0$ $\{1, 2, 7, 8, 10, 12\}$	0 0 28 86 35 44 59 24 40 62 68 77 70	9 2 00 00 00 00 58 38 00 73 00 73 00 00
$\iota_K P_0$ $\{3, 4, 7, 8, 10, 12\}$	0 0 25 53 35 44 59 24 40 62 68 77 70	0 0 49 59 00 00 58 38 00 73 00 73 00 00
$\iota_K P_0$ $\{1, 2, 3, 4, 7, 8, 10\}$	0 0 25 53 35 44 59 24 40 62 68 04 70	9 2 49 59 00 00 58 38 00 73 00 00 00 00
$\iota_K P_0$ $\{1, 2, 3, 4, 7, 8, 12\}$	0 0 25 53 35 44 59 24 40 49 68 77 70	9 2 49 59 00 00 58 38 00 00 00 73 00 00
$\iota_K P_0$ $\{1, 2, 3, 4, 7, 8, 10, 12\}$	0 0 25 53 35 44 59 24 40 62 68 77 70	9 2 49 59 00 00 58 38 00 73 00 73 00 00

TABLE S15. Certificate for Proposition 9.6(f), continued.

(i) The annihilator of Σ in reduced echelon form (blocks: $v_1, \dots, v_{13}$)

Υ_1	1 0 00 00 00 00 03 34 00 18 16 95 75	Υ_7	0 0 00 00 10 00 09 08 06 18 79 A8 35
Υ_2	0 1 00 00 00 00 03 99 60 A0 A8 57 30	Υ_8	0 0 00 00 01 00 04 78 49 20 58 5A AA
Υ_3	0 0 10 00 00 00 07 50 90 26 21 01 32	Υ_9	0 0 00 00 00 10 05 76 88 A8 07 46 38
Υ_4	0 0 01 00 00 00 09 59 61 23 58 37 18	Υ_{10}	0 0 00 00 00 01 05 8A 66 55 70 09 74
Υ_5	0 0 00 10 00 00 08 95 58 4A 06 16 29	Υ_{11}	0 0 00 00 00 00 13 46 84 20 31 01 47
Υ_6	0 0 00 01 00 00 0A 5A 76 74 33 84 34		

(ii) The points ρ_i over 1728 in $\mathbb{Q}_{121}$, their classes, and the positions of η_1, η_2

classes	ρ_1^*: 30, ρ_2: 85, ρ_3: 49, ρ_4^*: 90, ρ_5: 6A, ρ_6: 61, ρ_7: 00, ρ_8: 61, ρ_9: 6A, ρ_{10}: 42, ρ_{11}: 00, ρ_{12}: 86
$(\eta_{1,v}, \eta_{2,v})$	v_1: (ρ_1, ρ_4), v_2: (ρ_1, ρ_4), v_3: (ρ_{12}, ρ_6), v_4: (ρ_3, ρ_8), v_5: (ρ_6, ρ_{12}), v_6: (ρ_9, ρ_2), v_7: (ρ_8, ρ_3), v_8: (ρ_{11}, ρ_7), v_9: (ρ_8, ρ_3), v_{10}: (ρ_4, ρ_1), v_{11}: (ρ_2, ρ_9), v_{12}: (ρ_{10}, ρ_5), v_{13}: (ρ_{11}, ρ_7)
ℓ-adic logarithms	$(5, 9)$: 79, $(6, 8)$: 49, $(7, 11)$: 06

(iii) Region B, depth 1: the points of $\mathbb{Q}_{121}$ near each ρ_i, as $\text{class}_{\text{orbit}}$

$\bar{w} = 1$	ρ_1 $38_1\,33_1$; ρ_2 $49_2\,11_3$; ρ_3 $08_4\,8A_5$; ρ_4 $90_6\,90_7$; ρ_5 $7A_8\,5A_9$; ρ_6 $14_{10}\,09_{11}$; ρ_7 $66_{12}\,55_{13}$; ρ_8 $71_8\,51_9$; ρ_9 $17_{10}\,02_{11}$; ρ_{10} $81_5\,03_4$; ρ_{11} $65_{12}\,56_{13}$; ρ_{12} $42_2\,1A_3$
$\bar{w} = 2$	ρ_1 $50_1\,10_2$; ρ_2 $74_3\,96_4$; ρ_3 $1A_5\,78_6$; ρ_4 $90_7\,90_7$; ρ_5 $62_8\,67_9$; ρ_6 $88_{10}\,45_{11}$; ρ_7 $74_{12}\,47_{13}$; ρ_8 $64_9\,69_8$; ρ_9 $83_{10}\,46_{11}$; ρ_{10} $11_5\,73_6$; ρ_{11} $77_{12}\,44_{13}$; ρ_{12} $77_3\,95_4$
$\bar{w} = 3$	ρ_1 $37_1\,34_1$; ρ_2 $A3_2\,67_3$; ρ_3 $64_4\,23_5$; ρ_4 $90_6\,90_7$; ρ_5 $0A_8\,1A_9$; ρ_6 $35_{10}\,98_{11}$; ρ_7 $88_{12}\,33_{13}$; ρ_8 $01_8\,11_9$; ρ_9 $36_{10}\,93_{11}$; ρ_{10} $28_5\,67_4$; ρ_{11} $83_{12}\,38_{13}$; ρ_{12} $A8_2\,64_3$
$\bar{w} = 4$	ρ_1 $35_1\,36_1$; ρ_2 $58_2\,02_3$; ρ_3 $10_4\,77_5$; ρ_4 $90_6\,90_7$; ρ_5 $4A_8\,8A_9$; ρ_6 $56_{10}\,77_{11}$; ρ_7 $AA_{12}\,11_{13}$; ρ_8 $41_8\,81_9$; ρ_9 $55_{10}\,74_{11}$; ρ_{10} $74_5\,10_4$; ρ_{11} $A1_{12}\,1A_{13}$; ρ_{12} $53_2\,09_3$
$\bar{w} = 5$	ρ_1 $3A_1\,31_1$; ρ_2 $3A_2\,20_3$; ρ_3 $92_4\,A5_5$; ρ_4 $90_6\,90_7$; ρ_5 $2A_8\,AA_9$; ρ_6 $82_{10}\,40_{11}$; ρ_7 $99_{12}\,22_{13}$; ρ_8 $21_8\,A1_9$; ρ_9 $89_{10}\,40_{11}$; ρ_{10} $A6_5\,99_4$; ρ_{11} $92_{12}\,29_{13}$; ρ_{12} $31_2\,20_3$
$\bar{w} = 6$	ρ_1 $20_1\,40_2$; ρ_2 $30_3\,2A_4$; ρ_3 $84_5\,03_6$; ρ_4 $90_7\,90_7$; ρ_5 $66_8\,63_9$; ρ_6 $53_{10}\,7A_{11}$; ρ_7 $29_{12}\,92_{13}$; ρ_8 $68_9\,65_8$; ρ_9 $58_{10}\,71_{11}$; ρ_{10} $87_5\,08_6$; ρ_{11} $22_{12}\,99_{13}$; ρ_{12} $30_3\,21_4$
$\bar{w} = 7$	ρ_1 $80_1\,90_2$; ρ_2 $52_3\,08_4$; ρ_3 $61_5\,26_6$; ρ_4 $90_7\,90_7$; ρ_5 $68_8\,61_9$; ρ_6 $10_{10}\,02_{11}$; ρ_7 $A1_{12}\,1A_{13}$; ρ_8 $6A_9\,63_8$; ρ_9 $10_{10}\,09_{11}$; ρ_{10} $6A_5\,25_6$; ρ_{11} $AA_{12}\,11_{13}$; ρ_{12} $59_3\,03_4$
$\bar{w} = 8$	ρ_1 $A0_1\,70_2$; ρ_2 $63_3\,A7_4$; ρ_3 $A7_5\,90_6$; ρ_4 $90_7\,90_7$; ρ_5 $64_8\,65_9$; ρ_6 $A4_{10}\,29_{11}$; ρ_7 $38_{12}\,83_{13}$; ρ_8 $66_9\,67_8$; ρ_9 $A7_{10}\,22_{11}$; ρ_{10} $A4_5\,90_6$; ρ_{11} $33_{12}\,88_{13}$; ρ_{12} $68_3\,A4_4$
$\bar{w} = 9$	ρ_1 $32_1\,39_1$; ρ_2 $94_2\,76_3$; ρ_3 $36_4\,51_5$; ρ_4 $90_6\,90_7$; ρ_5 $9A_8\,3A_9$; ρ_6 $A3_{10}\,2A_{11}$; ρ_7 $77_{12}\,44_{13}$; ρ_8 $91_8\,31_9$; ρ_9 $A8_{10}\,21_{11}$; ρ_{10} $5A_5\,35_4$; ρ_{11} $74_{12}\,47_{13}$; ρ_{12} $97_2\,75_3$
$\bar{w} = 10$	ρ_1 $00_1\,60_2$; ρ_2 $41_3\,19_4$; ρ_3 $32_5\,55_6$; ρ_4 $90_7\,90_7$; ρ_5 $60_8\,69_9$; ρ_6 $37_{10}\,96_{11}$; ρ_7 $65_{12}\,56_{13}$; ρ_8 $62_9\,60_8$; ρ_9 $34_{10}\,95_{11}$; ρ_{10} $39_5\,56_6$; ρ_{11} $66_{12}\,55_{13}$; ρ_{12} $4A_3\,12_4$

TABLE S16. Certificate for Proposition 9.10(c), from the second program (Appendix B.2). $K = K_{288\text{a}1}$ has the places v_1, v_2 above 11 with $K_v = \mathbb{Q}_{11}$ and $v_3, \dots, v_{13}$ with $K_v = \mathbb{Q}_{121}$; classes are in $\mathbb{F}_{11}[t]/(T)$ (two digits, A = 10), in the coordinates of [4], and at v_1, v_2 only the first digit is used. The points marked * and the points of orbits of size one lie in $\mathbb{Q}_{11}$. The entry (i, i') gives, for ρ_i, $\rho_{i'}$ with the same class, the digits of $N_v(\text{Log}_v[\rho_i - \rho_{i'}]/121 \bmod 11)$, where N_v is the identity; then $v_{11}(\text{Log}_v[\rho_i - \rho_{i'}]) = 2$. The allowed configurations of region B at depth ≥ 2 are the choices, one point per place with pairwise distinct orbits, whose tuple of ramification points has its class tuple in Σ; each has $\beta = \eta_g$, S' the places where its ρ_i is not $\eta_{g,v}$, and ϑ^{f} built from these logarithms. For all 30720 of them the conditions of (c) hold, with the classes of (iii) for $R^{(1)}$.

S5. The exponent 29

The certificate of this section is used in the proof of Proposition 10.1.

	annihilator of Σ	j_1, orbit	classes in k_{v_1}, k_{v_2}
Υ_1^*	10000 00000 00304 52001 00216 13534	1, g_1	16106 34302 00312 00506 24433 06632
Υ_2^*	01000 00000 00000 14243 13010 31064	g_2	23314 43500 46230 40313 34165 40160
Υ_3^*	00100 00000 00301 62604 63011 15155	8, g_1	54206 36315 22521 64461 54036 54214
Υ_4	00010 00000 00601 31545 41103 00231	g_2	04031 13223 43510 64212 25523 14142
Υ_5	00001 00000 00605 64414 66015 30553	15, g_1	22306 31321 44030 51353 14332 32566
Υ_6	00000 10000 00201 56244 12520 44360	g_2	55455 53646 40160 11111 16251 51124
Υ_7	00000 01000 00105 45153 12316 42410	22, g_1	60406 33334 66246 45245 44635 10141
Υ_8	00000 00100 00000 51011 16343 40031	g_2	36102 23362 44440 35010 00616 25106
Υ_9^*	00000 00010 00006 15444 23164 12426	29, g_1	35506 35340 11455 32130 04231 65423
Υ_{10}	00000 00001 00504 40246 35260 15462	g_2	10526 63015 41020 52616 61344 62151
Υ_{11}	00000 00000 10402 30542 22211 53410	36, g_1	03606 30353 33664 26022 34534 43005
Υ_{12}	00000 00000 01404 40564 55314 62662	g_2	61243 33431 45300 06515 52002 36133
Υ_{13}	00000 00000 00011 41304 32016 11032	43, g_1	41006 32366 55103 13614 64130 21350
		g_2	42660 03154 42650 23414 43430 03115

Table S17. The disc test of Proposition 10.1(b) for $E = 864a1$, with the conventions of Table S11 for $\ell = 7$. Here $K = \mathbb{Q}[x]/(G_{29}(x, -216))$, and $k_{v_i} = \mathbb{F}_7[x]/(g_i)$ at the places v_1, v_2 above 7, where $g_1 = x^{15} + 6x^{12} + x^{11} + 6x^9 + x^8 + 3x^7 + 2x^5 + 3x^4 + 3x^3 + x^2 + 5x + 1$ and $g_2 = x^{15} + 6x^{14} + 3x^{13} + 5x^{12} + 5x^{11} + 6x^{10} + 5x^9 + 3x^8 + 4x^7 + 5x^4 + 6x^3 + 4x^2 + 4x + 4$ are the factors of $G_{29}(x, -216)$ modulo 7. A string $a_0 \dots a_{14}$ stands for $\sum_k a_k \nu^{7^k}$ with $\nu = x + 1$, and $\Upsilon(a \,|\, b) = \sum_k \upsilon_k a_k + \sum_k \upsilon_{15+k} b_k$. The row g_i of the disc of j_1 gives the classes $a_i \in k_{v_1}$ and $b_i \in k_{v_2}$ of a point of the Frobenius orbit whose x-coordinates reduce to the roots of g_i, at v_i the point Q with $x(Q) \equiv x$ modulo v_i, and at the other place the point whose residue has the lexicographically smallest coordinates in the basis $1, x, \dots, x^{14}$. Another choice there replaces the class by one of its Frobenius shifts, which the proof of Proposition 10.1 allows for. The rows marked * are used in the proof of Proposition 10.1.

S6. The exponent 25

Table S18 gives the eight forms of [3, Table 3] that we use, in the notation of Section 11.1, and the sextic fields of [3, Table 2]. Table S19 gives the fields L and the discriminants of L and $F = L(\zeta)$.

Table S20 gives the local data of Sections 14 and 15. The other tables of this section print, for each case of Part III, a basis of the image $\Sigma_{\mathfrak{P}} = \mathrm{loc}_{\mathfrak{P}}(\mathrm{Sel})$ of the Selmer group at the primes with $e(\mathfrak{P}) = 20$, and the classes ϑ (or $\mathrm{cl}(D_j)$) tested against it. With $r = [F_{\mathfrak{P}} : \mathbb{Q}_5]/2$ we use the basis $\kappa_{\mathfrak{P}}(G_1), \dots, \kappa_{\mathfrak{P}}(G_r), \kappa_{\mathfrak{P}}(T_\pi)$ of $W_{\mathfrak{P}}$ of Lemma 13.1, where $G_1, \dots, G_r$ are explicit points over $F_{\mathfrak{P}}$ (for $\mathcal{D}$ with $i' = 5$, points and two divisors of degree 2), listed with the Selmer bases as S-units in [4]. A vector of $W_{\mathfrak{P}}$ is written as its $r + 1$ coordinates in $\mathbb{F}_5$, in groups of five, with the coordinate of $\kappa_{\mathfrak{P}}(T_\pi)$ after a bar. The class $\mathrm{cl}_\nu(\gamma)$ of Section 13.2 is $(\overline{\pi^{-\nu} a_1}, \dots, \overline{\pi^{-\nu} a_r})$ modulo the last basis vector, printed with last coordinate 0. The basis of $\Sigma_{\mathfrak{P}}$ is in reduced row echelon form, and its last vector, $\kappa_{\mathfrak{P}}(T_\pi) = (0, \dots, 0 \,|\, 1)$ (for $i' = 24$,

i	h_i
5	$[-1, 1, 1, 1, -1, 5, -25/7, -35, -65, -215, 1025, -7975, -57025]$
13	$[-64, -64, -32, -16, -16, -32, -424/7, -76, -68, -28, 134, 859, 2207]$
6	$[3, 1, -2, 0, -4, -4, 24/7, 16, -80, -48, -928, -2176, 27072]$
23	$[192, 32, -32, 0, -16, -8, 24/7, 8, -20, -6, -58, -68, 423]$
16	$[-81, -27, -27, -27, -9, 9, 171/7, 33, 63, 141, 149, -67, -1657]$
14	$[-25, -50, -25, -10, -5, -10, -235/7, -50, -49, -34, 31, 614, 1763]$
22	$[100, 125, 50, 15, 0, -15, -270/7, -45, -36, -27, -54, -297, -648]$
24	$[-395, -153, -92, -26, 24, 40, 304/7, 48, 64, 64, 0, -128, -512]$
i'	defining polynomial of K, in a variable U
5	$U^6 + 10U^3 + 24U + 5$
6	$U^6 - 2U^5 - 6U - 3$
16	$U^6 + 10U^3 - 15U^2 + 18U - 10$
22	$U^6 + 3U^5 - 10U^3 + 12U - 4$
24	$U^6 - 10U^3 - 6U + 5$

TABLE S18. The forms h_i, $i \in \mathcal{I}$, and the fields K of the forms $h_{i'}$.

i'	defining polynomial of L, in a variable U	$\lvert \operatorname{disc} L \rvert$	$\lvert \operatorname{disc} F \rvert$
22	$U^{12} - 3U^{11} - 6U^{10} + 15U^9 + 30U^8 - 39U^7 - 58U^6 + 39U^5 + 30U^4 - 15U^3 - 6U^2 + 3U + 1$	$2^8 3^{14} 5^{11}$	$2^{32} 3^{56} 5^{52}$
6	$U^{12} - 4U^{11} - 8U^{10} + 30U^9 - 72U^7 + 108U^6 - 24U^5 - 120U^4 + 130U^3 - 64U^2 + 16U - 3$	$2^{18} 3^8 5^{11}$	$2^{72} 3^{32} 5^{52}$
16	$U^{12} - 6U^{11} + 36U^{10} - 20U^9 + 60U^8 + 732U^7 - 82U^6 - 1338U^5 + 180U^4 + 260U^3 + 6U^2 + 24U + 1$	$2^{18} 3^{14} 5^{11}$	$2^{72} 3^{56} 5^{52}$
24	$U^{12} - 36U^{10} - 120U^9 - 180U^8 - 144U^7 - 120U^6 + 144U^5 + 540U^4 + 720U^3 + 144U^2 - 144$	$2^{18} 3^{14} 5^{11}$	$2^{72} 3^{56} 5^{52}$
5	$U^{12} - 6U^{11} - 18U^{10} + 140U^9 + 180U^8 - 1818U^7 + 738U^6 + 7614U^5 - 7020U^4 - 11260U^3 + 12066U^2 + 5274U - 6143$	$2^{18} 3^{14} 5^{11}$	$2^{72} 3^{56} 5^{52}$

TABLE S19. The fields $L = K(\theta_1)$ and the absolute discriminants of L and of $F = L(\zeta)$. The root discriminant of F is about 32.7 for $i' = 22$, 33.6 for $i' = 6$ and 58.3 otherwise.

$(0, \dots, 0 \,|\, 1 \;\; 0, \dots, 0 \,|\, 1))$, is not printed. We also give $\kappa_{\mathfrak{P}}(P_0 - \infty)$ (for $\mathcal{D}$, $\kappa'_{\mathfrak{P}}(P'_0 - \infty)$), which lies in $\Sigma_{\mathfrak{P}}$. Each non-membership statement can be checked by reducing the printed class against the basis.

curve	i'	$[F_\mathfrak{P} : \mathbb{Q}_5]$	$\dim W_\mathfrak{P}$	$\operatorname{rank}\Sigma_\mathfrak{P}$	family	ν_1	$\operatorname{lev}(\gamma_n)$, $n \geq 2$	ε	$w - w(t_0)$	$x - x_0$	$v(y_0)$	$y - y_0$	Ω
$\mathcal{C}$	6	40	21	8	$t = 25s$	4	9, 13, 18, 19	34	≥ 14	≥ 14	4	≥ 10	≥ 20
$\mathcal{C}$	16	40	21	3	$t = 25s$	4	8, 12, 16, 16	30	≥ 10	≥ 10	0	≥ 10	≥ 20
$\mathcal{C}$	24	20	11	8 (9 jointly)	$t = 25s$	3	≥ 7	30	≥ 10	≥ 10	0	≥ 10	≥ 20
$\mathcal{C}$	22	40	21	6	$t = 5j + 25s$	1	Prop. 15.5	30	≥ 10	≥ 10	0	≥ 30	≥ 20
$\mathcal{D}$	22	40	21	8	$t = 25s$	3	7	30	≥ 10	≥ 10	0	≥ 30	≥ 20
$\mathcal{D}$	5	40	21	4	$t = 25s$	3	7	30	≥ 10	≥ 10	0	≥ 30	≥ 20

TABLE S20. Left: data at the primes $\mathfrak{P}$ of F above 5 with $e(\mathfrak{P}) = 20$. Right: Gauss valuations in $v = v_\mathfrak{P}$ on the disc $\{v(s) \geq -10\}$ for the families, in the local models, with local coordinates x, y and centre (x_0, y_0); Ω is the integrand, $j = 1, \ldots, 4$ for $i' = 22$ on $\mathcal{C}$, and for $i' = 24$ the values are those at $\mathfrak{P}_1$ and at $\mathfrak{P}_2$. Write $\log[P_{t(s)} - P_0] = \sum \gamma_n s^n$ (for $\mathcal{D}$, along the lifted family of Section 15.3). The levels $\operatorname{lev}(\gamma_n)$ are listed for $n = 2, \ldots, 5$ when $i' = 6, 16$ and for $n = 2$ otherwise, and $\nu_1 = \operatorname{lev}(\gamma_1)$, except for $\mathcal{C}$ with $i' = 22$, where ν_1 is the level of the centre divisors D_j of Proposition 15.5. Also $\operatorname{rank}\Sigma_\mathfrak{P} = \dim \mathrm{Sel}$ (for $\mathcal{D}$, $\dim \mathrm{Sel}(J_\mathcal{D})$), except for $i' = 24$, where only the pair of primes gives an injective map. In every case $F_\mathfrak{P} = L_{\mathfrak{p}_L}(\zeta)$ for the prime $\mathfrak{p}_L$ of L below $\mathfrak{P}$, with $[F_\mathfrak{P} : L_{\mathfrak{p}_L}] = 4$ (2 for $i' = 24$), $[F_\mathfrak{P} : M_{\mathfrak{p}_M}] = 2$ for $i' \in \{5, 22\}$, $\kappa_\mathfrak{P}(T_\pi) \neq 0$ (for $\mathcal{D}$, $\kappa'_\mathfrak{P}(T'_\pi) \neq 0$) and $5\mathcal{O}^2_\mathfrak{P} \subseteq \Lambda$. For $\mathcal{D}$ the column $w - w(t_0)$ also bounds $Z - 1$, and $(Y - \sqrt{\beta})/\xi - 1$ has Gauss valuation ≥ 30. For $i' = 24$ the values at $\mathfrak{P}_2$ are those at $\mathfrak{P}_1$, since σ exchanges $\mathfrak{P}_1$ and $\mathfrak{P}_2$ and the family has coefficients in L. In every row ε exceeds $5e(\mathfrak{P})/4 = 25$ and $y - y_0$ exceeds $v(y_0)$, so Lemmas 13.5 and 13.7 apply.

Selmer basis	10000 00401 41023 14211	0
	01000 00410 41023 12333	0
	00100 00314 12432 22333	0
	00010 00201 41420 11411	0
	00001 00024 30440 00100	0
	00000 10002 41311 13111	0
	00000 01323 04420 02023	0
ϑ, level 4	34114 00233 41424 12100	0
$\kappa_\mathfrak{P}(P_0 - \infty)$	30040 30003 00022 00000	2

TABLE S21. $\mathcal{C}$, $i' = 6$, $r = 20$: $\dim \Sigma_\mathfrak{P} = 8$ (Theorem 14.1).

Selmer basis	13044 30414 11014 00201	0
	00132 23421 00330 22404	0
ϑ, level 4	02444 11022 31324 10200	0
$\kappa_\mathfrak{P}(P_0 - \infty)$	13121 03330 11344 22100	0

TABLE S22. $\mathcal{C}$, $i' = 16$, $r = 20$: $\dim \Sigma_\mathfrak{P} = 3$ (Theorem 14.1).

Selmer basis	10000 00440 | 0	00104 32140 | 4
	01000 00043 | 0	02022 13142 | 4
	00100 00441 | 0	04404 22423 | 2
	00010 00143 | 0	02430 23304 | 4
	00001 00001 | 0	04300 40434 | 1
	00000 10210 | 0	00431 11141 | 3
	00000 01223 | 0	00044 01102 | 3
	00000 00000 | 0	11401 11214 | 3
$(\vartheta_1, \vartheta_2)$, level 3	00001 00000 | 0	30000 00000 | 0
$\kappa_{\mathcal{P}}(P_0 - \infty)$	11410 13404 | 3	44104 44341 | 2

TABLE S23. $\mathcal{C}$, $i' = 24$, at $\mathfrak{P}_1$ then $\mathfrak{P}_2$, $r = 10$ each: $\dim \Sigma_{\mathcal{P}} = 9$ (Theorem 14.2).

Selmer basis	10000 11411 30104 11041 | 0
	01000 31233 33443 03332 | 0
	00100 23340 13020 24000 | 0
	00010 14133 41320 01100 | 0
	00001 20424 21330 14041 | 0
$\mathrm{cl}(D_1)$	34042 01341 21133 30441 | 0
$\mathrm{cl}(D_2)$	43034 02132 40211 11332 | 0
$\mathrm{cl}(D_3)$	32021 03423 12344 43223 | 0
$\mathrm{cl}(D_4)$	01013 04214 32422 21114 | 0
$\kappa_{\mathfrak{P}}(P_0 - \infty)$	22314 32312 20034 22200 | 3

TABLE S24. $\mathcal{C}$, $i' = 22$, $r = 20$: $\dim \Sigma_{\mathfrak{P}} = 6$, and $\mathrm{cl}(D_j)$ at level 1 (Proposition 15.5).

Selmer basis	10000 04200 13101 13300 | 0
	01000 00000 44121 34400 | 0
	00100 02000 21311 00000 | 0
	00010 02304 43114 30300 | 0
	00001 01203 43113 42300 | 0
	00000 11404 42434 31200 | 0
	00000 00010 01314 40100 | 0
ϑ, level 3	20000 00000 00000 00000 | 0
$\kappa'_{\mathfrak{P}}(P'_0 - \infty)$	21200 41041 14202 34200 | 2

TABLE S25. $\mathcal{D}$, $i' = 22$, $r = 20$: $\dim \Sigma'_{\mathfrak{P}} = 8$ (Theorem 15.6).

Selmer basis	10003 00240 34304 20400 | 0
	01031 30131 01003 40100 | 0
	00111 24103 43034 20200 | 0
ϑ, level 3	40000 00000 00000 00000 | 0
$\kappa'_{\mathfrak{P}}(P'_0 - \infty)$	30443 31022 04423 40000 | 3

TABLE S26. $\mathcal{D}$, $i' = 5$, $r = 20$: $\dim \Sigma'_{\mathfrak{P}} = 4$ (Theorem 15.7).

S7. Defining polynomials of the fields K_E

For the fields K_E with $p \geq 17$ other than those printed in Appendix C.2, Tables S27, S28, S29 and S30 give defining polynomials F_E with smaller coefficients than $G_p(x, j_E)$ and $K_E \simeq \mathbb{Q}[x]/(F_E)$ (Proposition 2.8(a)); the isomorphism can be checked with PARI/GP's `nfisisom`.

k	F_{864a1}	F_{864b1}	F_{864c1}
18	1	1	1
17			−6
16			−51
15	−170	−170	136
14			1836
13	204	816	816
12	12 767	12 767	−29 172
11			−80 376
10	−18 768	−75 072	132 294
9	−437 580	−437 580	921 876
8	10 404	166 464	922 590
7	895 560	3 582 240	−2 398 632
6	5 506 351	5 506 351	−4 025 940
5	902 088	14 433 408	7 789 536
4	−10 111 056	−40 444 224	18 893 868
3	10 142 710	14 769 430	−4 375 528
2	294 372	4 709 952	−19 282 335
1	7260	355 632	30 729 954
0	17	17	47 607 325

Table S27. Defining polynomials $F_E = \sum_k c_k x^k$ of the fields $K_E \simeq \mathbb{Q}[x]/(F_E)$ for $p = 17$ (coefficients c_k; empty cells are 0).

k	F_{54a1}	F_{96a1}	F_{864a1}	F_{864b1}	F_{864c1}
20	1	1	1	1	1
19	−5	−8	−6	−4	−4
18	−95			−38	−38
17	1311		−228	152	152
16	−7866	228	−114	475	475
15	29 868			−2128	−2128
14	−84 208	1710	38	−494	−494
13	196 916	−2736	−684	11 780	11 780
12	−400 558	6840		−38 855	−38 855
11	705 546	−4560	2204	5168	5168
10	−1 095 654	24 624	−1368	321 328	321 328
9	1 592 010	−21 888		−77 368	−369 208
8	−2 180 516	122 949	3553	−1 219 648	−635 968
7	2 680 444	−24 168	−342	−4 825 088	1 887 232
6	−3 094 568	120 840		10 643 648	−2 780 992
5	3 565 236	−151 392	2432	42 433 536	−4 844 544
4	−3 681 003	467 628	1026	−101 180 928	15 846 912
3	3 121 491	−372 096		−105 762 816	7 179 264
2	−2 622 627	454 176	1216	318 708 736	−28 580 864
1	2 080 431	−203 520	336	−185 632 768	−5 552 128
0	−1 739 466	202 704	36	692 772 352	19 436 032

TABLE S28. Defining polynomials $F_E = \sum_k c_k x^k$ of the fields $K_E \simeq \mathbb{Q}[x]/(F_E)$ for $p = 19$ (coefficients c_k; empty cells are 0).

k	F_{864a1}	F_{864b1}	F_{864c1}	F_{27a1}
24	1	1	1	1
23		−6	−12	
22			120	
21	−184	−184	−644	−92
20	−1104	345	4209	
19	552	3036	−21 390	
18	56 304	15 272	83 536	2530
17	616 860	25 806	−357 420	
16	4 164 702	−206 034	1 219 920	
15	21 648 152	−1 175 898	−3 906 734	20 332
14	91 128 024	−3 447 792	11 706 954	
13	317 580 228	488 244	−17 672 142	
12	927 022 820	37 588 325	7 650 789	85 629
11	2 255 786 088	154 470 300	86 638 884	
10	4 528 964 040	290 725 704	−406 905 834	
9	7 369 449 964	−34 649 454	253 390 310	180 320
8	9 393 674 973	−2 083 640 610	330 115 113	
7	9 012 675 672	−7 897 107 270	−1 375 571 580	
6	6 037 849 048	−18 074 010 880	5 863 441 060	126 592
5	2 590 035 060	−32 795 655 564	9 742 703 400	
4	1 480 144 668	−55 135 653 015	−13 074 470 520	
3	1 910 081 184	−90 410 641 672	−25 469 076 496	−47 104
2	2 702 447 928	−118 890 670 248	2 886 994 224	
1	2 547 351 672	−77 358 701 298	22 839 931 776	
0	1 175 980 708	−19 497 211 127	10 923 487 888	−94 208

TABLE S29. Defining polynomials $F_E = \sum_k c_k x^k$ of the fields $K_E \simeq \mathbb{Q}[x]/(F_E)$ for $p = 23$ (coefficients c_k; empty cells are 0).

k	F_{27a1}	F_{54a1}	F_{96a1}	F_{864a1}	F_{864b1}	F_{864c1}
30	1	1	1	1	1	1
29		−3	−8	−12	−6	−6
28		−174			−87	−87
27		2030		−1856	580	580
26		−7569	696	−1392	2262	2262
25		−16 965	−2784		−21 576	−21 576
24	−348	275 384	1044	82 012	9048	9048
23		−1 026 948	−5568	1218	349 044	349 044
22		−133 023	119 712	2088	−1 348 239	−1 348 239
21		20 953 805	−434 304	−816 640	−690 374	−690 374
20		−131 145 018	835 896	386 106	19 770 315	19 324 875
19		504 570 594	1 241 664	85 608	−43 634 328	−54 324 888
18	54 462	−1 324 685 577	14 950 254	4 675 670	−46 169 305	−11 424 985
17		2 198 703 411	−58 335 936	−4 076 646	236 978 082	561 703 842
16		−1 318 558 428	29 823 948	1 536 768	−476 453 151	−1 600 298 271
15		−4 025 629 432	−76 273 248	−18 109 920	−1 578 383 580	1 141 918 500
14		11 921 554 059	817 554 312	13 653 084	−11 790 818 811	8 153 757 189
13		2 140 486 143	−3 099 268 512	−2 689 344	44 918 002 386	−29 605 891 374
12	−2 628 908	−94 618 826 466	12 354 208 452	2 503 164	−134 507 492 165	48 856 992 955
11		250 158 780 882	8 417 454 624	2 881 614	−205 667 109 816	20 010 611 784
10		−323 068 359 627	−38 285 403 048	11 457 900	1 552 541 363 175	−251 359 677 465
9		364 322 778 905	−387 065 898 144	121 381 472	−5 899 843 004 438	693 796 849 642
8		−167 964 645 744	788 137 300 584	−170 156 166	15 265 431 142 077	−835 905 910 083
7		340 520 631 228	1 841 284 859 904	114 290 856	−44 320 220 770 476	179 235 229 524
6	3 287 817	−3 234 974 952 557	−1 484 875 544 391	76 075 265	131 970 490 813 968	2 563 378 744 848
5		5 214 582 572 727	−14 764 680 631 512	−191 272 110	−341 667 616 692 216	−6 264 435 390 456
4		18 351 950 771 082	16 548 120 296 316	133 535 952	687 954 645 137 598	11 034 458 175 678
3		39 292 459 404 830	33 403 947 359 040	−38 338 464	−1 008 635 348 809 004	−12 066 970 625 324
2		61 288 772 551 653	−11 264 566 705 488	−13 485 696	1 042 769 317 444 413	11 767 791 172 413
1		26 297 011 022 457	−145 786 684 685 184	10 583 856	−667 427 895 320 550	−6 366 521 947 110
0	−1 353 024	−12 531 251 052 076	177 841 423 987 200	2 476 408	187 473 369 342 293	3 527 041 923 413

TABLE S30. Defining polynomials $F_E = \sum_k c_k x^k$ of the fields $K_E \simeq \mathbb{Q}[x]/(F_E)$ for $p = 29$ (coefficients c_k; empty cells are 0).

S8. The computational statements

Table S31 lists the statements of the paper whose proofs rest on machine computation, with the computation behind each. Appendix B explains why the results are exact or carry proven error bounds, and which computations were repeated.

Table S31: The computational statements of the proofs.

Statement	What is computed
Part I	
Section 2.3, Lemma 2.5, Table 5, Appendix C.1	the j-maps (2.2) and (2.3) from q-expansions, and their check against the q-expansion of j; for $p = 23$, 29 the model $y^2 = F(x)$ and the polynomials $\mathcal{A}_p$, $\mathcal{B}_p$, f_0, f_1 by exact linear algebra, the identities to high order, and $f_0^2 - f_1^2 F$; the fibre polynomials G_p; the fibres over 0 and 1728; the irreducibility of $\mathcal{B}_{19}$ (Lemma 2.11)
Proposition 2.8, Tables 6, 7 and 8, Propositions 2.9 and 8.5	the fields K_E: irreducibility of $G_p(x, j_E)$, discriminants, maximal orders, signatures, splitting of ℓ, p and other primes, subfields and automorphisms, isomorphism with $\mathbb{Q}[X]/\Phi_p(X, j(E))$ or [2, Table 5]; the Frobenius elements of Proposition 2.8(d); for $K_{288\mathrm{a}1}$ the defining polynomial from the subgroups of order p and its certificate (a factor of the p-division polynomial and its closure under multiplication maps), the roots of H_{10} in K and their valuations, and (Proposition 8.5) the completions at the places above 2 and 3, the factorization $G_{19}(x, 1728) = H_{10}^2$ and H_{10} mod 3; the class groups for $p = 23$, 29 under GRH
Proposition 2.2	the seven CM j-invariants
Proposition 2.3 (a repetition of the computations of [2, §7.1])	the fields K_α and L_α of Appendix B.2 and the class numbers of K_α and L_α; the 2-Selmer groups of the four curves E_c over K_α; their torsion; the saturation of explicit points at the odd primes $\ell \leq 997$; the lower bound for the canonical height and the index bounds; the rational points of 121b1; the Mordell–Weil sieves and the p-adic linear terms of the elliptic Chabauty step; the j-invariants at the seven rational points of 121b1 over $x = \infty$, 4, 5/4, -2
Lemma 9.1, Table 10	$a_\ell(E)$; the traces of all elliptic curves over $\mathbb{F}_\ell$
Section 2.6, Lemma 2.15	isogeny classes and j-invariants; the irreducibility of $G_p(x, -12288000)$; the integral points on $y^2 = x^3 \pm 1$
Sections 3.5 and 3.6	Hecke data at level p; $\#\widetilde{J}(\mathbb{F}_{11})$, $\#\widetilde{J}(\mathbb{F}_{121})$, $\#\widetilde{J}(\mathbb{F}_{7^5})$
Propositions 3.11, 3.7 and 3.8	the Selmer groups: S-unit bases, square, fifth-power and cube classes at the places above ℓ and p, local images spanned by explicit points, linear algebra over $\mathbb{F}_\ell$
Part II	
Propositions 7.2, 7.6, 7.7, 6.2, 8.1 and 8.6	the branch conditions of Section 4.1 in the fields $\mathbb{Q}(\xi)$; the roots of the representative fibres in each K_v; their classes; the tests against Σ
Lemma 7.3, Propositions 7.5 and 8.4	the expansions of $1/j$ at the cusps; the points near the cusps; their classes
Propositions 7.6 and 7.7, Proposition 8.3	the lattices Λ_v and groups Θ_v; $\Sigma \cap \Theta$; the hypotheses of Lemmas 5.4 and 5.8; the conjugates of the known points; the levels and classes in all cases
Lemma 8.8, Propositions 8.9 and 8.10	the branch conditions at the roots of H_{10} for $\kappa = \pm 3^7, \pm 3^9$; the centres at the places above 3; the lattices Λ_v and groups Θ_v; $\Sigma \cap \Theta$; the levels and classes in all 53184 and 44928 cases
Proposition 7.7 (sieve), Propositions 6.3 and 8.7	the sieves of Lemma 5.11 over finite fields; for Proposition 8.7, the closed points of the reduced fibres with their degrees and ramification indices
Proposition 6.4	the facts (K1)–(K4)

Statement	What is computed
Proposition 9.2, Proposition 10.1	$K(S, \ell)$ and its localization; Σ, Θ_T; the separability of $G_p(x, j_D)$ at the representatives j_D of all disc tests at $p = 23, 29$, including those of the ramified regions; the class tuples of all admissible configurations over the representatives; the tangent classes ϑ of Lemma 5.9
Propositions 9.6 and 9.10	the fibres in the ramified regions; the disc tests by meet-in-the-middle; the sets S_0 and the tests (d), (e); the 11-adic logarithms of the fixed terms Γ_v; the maps N_v; the components π_v at the points over 0 and 1728 (Lemma 9.4); the sieves at 43 and 89
Proposition 10.2, Proposition 10.1(d)	the disc test at depth 1 of region A against $\Sigma + \Theta_T$
Part III	
Lemma 11.8, Table 12	the three identities of forms and the correspondence of the factors of degree 10 over K; $f_i^2 + g_i^3 + h_i^5 = 0$ for the eight forms; the values in the last column of Table 12
Lemmas 12.1 and 12.3, Table S19	the factorization of $h_{i'}$ over K; the field L with the embedding of K; the identity $H = A\ell_1^{10} + B\ell_1^5\ell_2^5 + C\ell_2^{10}$; $B^2 = -121AC$; the point P_0 on $\mathcal{C}$; the integrality of β
Tables 13 and S20	the fields $F = L(\zeta_5)$: $[F : L] = 4$, signature $(0, 24)$, discriminants; the primes of L, M and F above 5 and their local degrees
Propositions 12.6 and 15.2	the units and S-units of F; the local images $W_{\mathfrak{p}}$; the Selmer groups; the independence of the Selmer bases
Proposition 12.6, Lemma 12.5	the Selmer groups over L and M from the bases of Proposition A.13(c); the local dimensions entering the Greenberg–Wiles differences; the eigenspace dimensions of the Selmer groups over F
Lemma 11.2	the signatures, contents n_i and decomposition of 5 in the fields K; the reductions of h_i modulo 5 and $v_5(h_i) = 2$ at r_i; the resultants of q_i and H_i; the local computation at 2 and 3; the values of q_i and H_i at the simple roots modulo 5; the fifth-power residue characters of the units at the primes above 11, 31, 41, 61, 71, 101
Lemma 11.3	the factorizations of the forms h_i over the sextic fields K; the units of Proposition A.13(b); the sieve over the 125 unit classes at the primes 11, 31, 101 and at the primes above 5 (all primes $7 \le p < 1000$ give the same sets); the decompositions $H_i(u_0, v_0) = \varepsilon_0\beta^5$
Proposition 11.4	for Stoll's curve D_0: the fake 2-Selmer set over k with the local conditions at 2, 3, 29; the 2-Selmer group of the elliptic curve E over k (dimension 6); the 2-adic computation on the two residue discs
Lemma 11.5	the 2-Selmer groups of the Jacobians of $y^2 = x^5 + 2^8 5^3$ and $y^2 = x^5 + 2^6 5^3$ and bounds for their torsion; the fake 2-Selmer sets of the curves M_3, M_4 and $D_{\pm 1}$, $D_{\pm 2}$ of [3, §§7–8]; the 2-Selmer group of the Jacobian of M_4; $E(\mathbb{Q}) \simeq \mathbb{Z}/2$ for $y^2 = x^3 + x^2 - 83x + 88$; the orders 1240 and 70 of the reductions of $[(1, 12) - \infty]$ in $J(\mathbb{F}_7)$ and $J(\mathbb{F}_{11})$
proof of Theorem 11.6, (M1) and (M2) of Appendix B.2	for $C_1 \colon y^2 = x^5 + 2^8 5^3$: $\#J(\mathbb{F}_q)$ and the orders of $[(-4, 176) - \infty]$ for $q < 700$; the witnesses for saturation at 2, 3, 5, 7, 11, 19; the Mordell–Weil sieve modulo $2^3 \cdot 3^2 \cdot 5^2 \cdot 7 \cdot 11 \cdot 19$; the 7-adic integrals I_0, I_1 (Coleman integration and the tiny integral on $5P$) and the power series at ∞ and at $(-4, 176)$. For $\tilde{D}_{-1}$, $\tilde{D}_{-2}$: the local images at 2, 3, 5 and the 2-Selmer groups (dimension 2); the classes of $Q_{j,1}$, $Q_{j,2}$; the function with divisor $2Q_{j,1} - 3\infty_{\pm} - 5\infty_{\mp}$
Propositions 12.8 and 12.9	the refinement of $\mathbb{P}^1(\mathbb{Z}_p)$ into residue classes; the elements of Sel passing all tests
Proposition 15.1, Lemma 15.3	the identities of (b) and (c) as identities of rational functions in Y, $\sqrt{\beta}$ and ξ, given $X^5 = Y^2 - \beta$ and $Y - \sqrt{\beta} = \xi Z^5$; the curve $\mathcal{D}$, the point P_0' and the prime divisors of β'; the fact that τ fixes $\sqrt{5}$, β', $\sqrt{\beta'}$, ξ, P_0' and $\mathfrak{P}$

Statement	What is computed
Table S20, Theorems 14.1 and 14.2, Proposition 15.5, and Theorems 15.6 and 15.7	the generators of $W_{\mathfrak{P}}$ and their logarithms; the lattices Λ; the coefficients γ_n; the levels and classes; the image $\mathrm{loc}_{\mathfrak{P}}(\mathrm{Sel})$
Part IV	
Theorem 16.1, Section 16.2	the description of j in lines 3, 4, 5, 7 of [2, Table 2] for $p = 13$ over all (a, b) modulo 3^7; the irreducibility of G_{j_E}; the fields K_E, F, L_E, their maximal orders, discriminants and subfields; $\mathrm{Aut}(K_E)$
Proposition 16.4, Table 14, Tables S32 and S33	the S-units of L_E (PARI/GP 2.19.0) and the exact check of their valuations; the cubic residue characters; the local images W_v from explicit points; the Selmer groups; the known points and their classes; the relation among the known classes for 96a1
Proposition 16.5, Table S34	the bound B_E of [1, Corollary 2.2] from the decomposition of the primes below it; the prime ideals of norm below B_E and their certificates; the orders of the fixed ideals
Lemma 17.3, Proposition 17.4, Table S35	the branch conditions in the fields $\mathbb{Q}(\xi)$ for every leaf; the balls of v-values, the lift condition and the classes; the comparison with $\mathrm{Sel}^{\lambda}(J/K)$
Lemma 17.5, Propositions 17.7 and 17.9, Lemma 17.8, Table S36	the logarithms, lattices $\Lambda_{\mathfrak{p}}$ and groups $\Theta_{\mathfrak{p}}$ at the primes above 3; the centres and the branches through them; the levels and classes in all cases; $\#J_1(13)(\mathbb{F}_{79})$ and $\#J_1(13)(\mathbb{F}_{337})$; the classes of the cusps
Appendix A	
Appendix A, Tables 15 and 16, Proposition A.13	the maximal orders; the prime ideals of norm below the bounds X and their relations; the values of R; the S-unit and Hilbert symbol checks; the cubic, quintic and quadratic residue characters of the S-units; the Minkowski-bound certificates of Table 16

S9. The exponent 13

The tables of this section are used in the proofs of Propositions 16.4, 16.5, 17.4, 17.7 and 17.9.

E	q	(e, f) of v	(e, f) of the primes of L_E above v	h_v	$\dim W_v$
96a1	2	$(2, 1)$	$(4, 2)$	0	0
	2	$(4, 2)$	$(4, 2)$, $(4, 2)$, $(4, 2)$, $(4, 2)$	2	2
	2	$(2, 2)$	$(4, 2)$, $(4, 2)$	0	0
	3	$(13, 1)$	$(26, 2)$	0	13
	3	$(1, 1)$	$(2, 2)$	0	1
	13	$(13, 1)$	$(13, 1)$, $(13, 1)$, $(26, 1)$	1	1
	13	$(1, 1)$	$(1, 1)$, $(1, 1)$, $(2, 1)$	1	1
864a1	2	$(2, 1)$	$(4, 2)$	0	0
	2	$(2, 2)$	$(4, 2)$, $(4, 2)$	0	0
	2	$(4, 2)$	$(4, 2)$, $(4, 2)$, $(4, 2)$, $(4, 2)$	2	2
	3	$(6, 1)$	$(6, 1)$, $(6, 1)$, $(6, 1)$, $(6, 1)$	2	8
	3	$(2, 1)$	$(2, 2)$, $(2, 2)$	1	3
	3	$(3, 2)$	$(6, 2)$, $(6, 2)$	1	7
	13	$(14, 1)$	$(28, 1)$, $(28, 1)$	0	0
864b1	2	$(2, 1)$	$(4, 2)$	0	0
	2	$(4, 2)$	$(4, 2)$, $(4, 2)$, $(4, 2)$, $(4, 2)$	2	2
	2	$(2, 2)$	$(4, 2)$, $(4, 2)$	0	0
	3	$(6, 1)$	$(6, 2)$, $(6, 2)$	1	7
	3	$(2, 1)$	$(2, 1)$, $(2, 1)$, $(2, 1)$, $(2, 1)$	2	4
	3	$(3, 2)$	$(6, 2)$, $(6, 2)$	1	7
	13	$(1, 1)$	$(2, 1)$, $(1, 2)$	0	0
	13	$(13, 1)$	$(26, 1)$, $(13, 2)$	0	0
864c1	2	$(2, 1)$	$(4, 2)$	0	0
	2	$(2, 2)$	$(4, 2)$, $(4, 2)$	0	0
	2	$(4, 2)$	$(4, 2)$, $(4, 2)$, $(4, 2)$, $(4, 2)$	2	2
	3	$(2, 1)$	$(2, 1)$, $(2, 1)$, $(2, 1)$, $(2, 1)$	2	4
	3	$(6, 1)$	$(6, 2)$, $(6, 2)$	1	7
	3	$(3, 2)$	$(6, 2)$, $(6, 2)$	1	7
	13	$(1, 1)$	$(2, 1)$, $(1, 2)$	0	0
	13	$(13, 1)$	$(26, 1)$, $(13, 2)$	0	0

TABLE S32. The places v of K_E above $q = 2$, 3, 13, the primes of L_E above them, $h_v = \dim J(K_v)[\lambda]$ and the dimension of the local image W_v (see (16.2)). For every curve $\sum_v \dim W_v = \dim L(S, 3)^-$. The local images are spanned by the classes of explicit K_v-points.

E	generators	residue characters	$\dim L(S, 3)$	$+ \,/\, -$ parts	$\dim \mathrm{Sel}$	rank at 3
96a1	43	43	43	25 / 18	4	4
864a1	45	45	45	25 / 20	2	2
864b1	47	47	47	27 / 20	4	4
864c1	47	47	47	27 / 20	2	2

TABLE S33. The S-unit bases of L_E ($\#S_L + 28$ generators), the number of independent cubic residue characters at auxiliary primes of degree 1 (equal to the number of generators, so they are independent modulo cubes), $\dim L(S, 3)$ and the dimensions of its τ-invariant and anti-invariant parts, $\dim \mathrm{Sel}^\lambda(J/K)$, and the rank of its restriction to $\bigoplus_{v|3} W_v$ (Proposition 16.4).

E	B_E	prime ideals certified	fixed ideals	classes	$\#\,\mathrm{Cl}(L_E)$
96a1	3433	517 (norm $\le B_E$)	a prime of norm 7	257, 260	divides 2
864a1	4931	685 (norm $< B_E$)	three ideals $\mathfrak{g}_1, \mathfrak{g}_2, \mathfrak{g}_3$	all 32 classes occur	divides 32
864b1	4888	613 (norm $< B_E$)	PARI's generator	333, 280	divides 2
864c1	5171	665 (norm $\le B_E$)	a prime of norm 3	349, 316	divides 2

TABLE S34. The certificates of Proposition 16.5. Every prime ideal of L_E of the stated norms has an exact certificate $\mathfrak{P} \cdot \mathfrak{g}^e = (\alpha)$ with respect to the fixed ideals, with α given explicitly, in factored form except for 864c1; for 864b1 the fixed ideal $\mathfrak{g}$ is the generator returned by PARI, which is not prime, and a separate certificate puts a prime of norm 3 in its class. "Classes" gives the numbers of prime ideals in the trivial class and in the class of $\mathfrak{g}$. Under GRH the prime ideals of norm less than B_E generate $\mathrm{Cl}(L_E)$ [1, Corollary 2.2], and the last column follows from certificates that $\mathfrak{g}^2$ is principal; for 864a1, that $\mathfrak{g}_1^4$, $\mathfrak{g}_2^4$, $\mathfrak{g}_3^2$ are principal, from certificates that $\mathfrak{P}_{19}^4$, $\mathfrak{P}_{13}^4$, $\mathfrak{P}_{283}^2\mathfrak{P}_{13}^2$ are principal. For 864a1 the column "classes" says that the exponent vectors of the certificates take all 32 values in $\mathbb{Z}/4 \times \mathbb{Z}/4 \times \mathbb{Z}/2$.

Table S35: The leaves of Proposition 17.4: the numbers of classes in $D_\mathfrak{p}(\mathcal{L})$ at the primes $\mathfrak{p} \mid 3$ in the order of Table S32 ("all" means all of $W_\mathfrak{p}$), and the number of elements of $\mathrm{Sel}^\lambda(J/K)$ whose restriction lies in $\prod_\mathfrak{p} D_\mathfrak{p}(\mathcal{L})$; a leaf with 0 is excluded.

E	leaf	$\#D_\mathfrak{p}(\mathcal{L})$	compatible
96a1	$t \equiv 0 \pmod 9$	1, 1	1
	$t \equiv 3 \pmod 9$	1, 1	1
	$t \equiv 6 \pmod 9$	1, 1	1
	$t \equiv 1 \pmod 9$	2, 1	0
	$t \equiv 4 \pmod 9$	2, 1	0
	$t \equiv 7 \pmod 9$	2, 1	0
	$t \equiv 2 \pmod 9$	2, 1	0
	$t \equiv 5 \pmod 9$	2, 1	0
	$t \equiv 8 \pmod 9$	2, 1	0
864a1	$-216 + 3^7\mathbb{Z}_3$	4, 2, 2	2
	$513 + 3^7\mathbb{Z}_3$	4, 2, 2	0
	$1242 + 3^7\mathbb{Z}_3$	4, 2, 2	0
	$-108 + 3^6\mathbb{Z}_3$	4, 2, 2	0
	$135 + 3^6\mathbb{Z}_3$	4, 2, 2	0
	$378 + 3^6\mathbb{Z}_3$	4, 2, 2	0
	$-27 + 3^6\mathbb{Z}_3$	4, 2, 2	0
	$216 + 3^6\mathbb{Z}_3$	4, 2, 2	0
	$459 + 3^6\mathbb{Z}_3$	4, 2, 2	0
864b1	$27 + 3^7\mathbb{Z}_3$	4, 2, 2	2
	$756 + 3^7\mathbb{Z}_3$	4, 2, 2	2
	$1485 + 3^7\mathbb{Z}_3$	4, 2, 2	2
	$729 + 3^8\mathbb{Z}_3$	4, 2, 4	0
	$-729 + 3^8\mathbb{Z}_3$	4, 2, 4	0
	$1458 + 3^8\mathbb{Z}_3$	4, 2, 4	0
	$-1458 + 3^8\mathbb{Z}_3$	4, 2, 4	0
	$19683 + 3^{11}\mathbb{Z}_3$	4, 2, 4	0

E	leaf	$\#D_{\mathfrak{p}}(\mathcal{L})$	compatible
	$-19683 + 3^{11}\mathbb{Z}_3$	4, 2, 4	0
	$39366 + 3^{11}\mathbb{Z}_3$	4, 2, 4	0
	$-39366 + 3^{11}\mathbb{Z}_3$	4, 2, 4	0
	$\{3^{12}t^3\}$	4, all, 4	0
	$\{2 \cdot 3^{12}t^3\}$	4, all, 4	0
864c1	$27 + 3^7\mathbb{Z}_3$	2, 4, 2	0
	$756 + 3^7\mathbb{Z}_3$	2, 4, 2	0
	$1485 + 3^7\mathbb{Z}_3$	2, 4, 2	0
	$729 + 3^8\mathbb{Z}_3$	2, 4, 4	0
	$-729 + 3^8\mathbb{Z}_3$	2, 4, 4	0
	$1458 + 3^8\mathbb{Z}_3$	2, 4, 4	0
	$-1458 + 3^8\mathbb{Z}_3$	2, 4, 4	0
	$19683 + 3^{11}\mathbb{Z}_3$	2, 4, 4	0
	$-19683 + 3^{11}\mathbb{Z}_3$	2, 4, 4	0
	$39366 + 3^{11}\mathbb{Z}_3$	2, 4, 4	0
	$-39366 + 3^{11}\mathbb{Z}_3$	2, 4, 4	0
	$\{3^{12}t^3\}$	all, 4, 4	0
	$\{2 \cdot 3^{12}t^3\}$	all, 4, 4	0

E, disc	$\dim \mathrm{loc}(\mathrm{Sel})$	$\dim \Theta$	$\dim(\mathrm{loc}(\mathrm{Sel}) + \Theta)$	cases, excluded
864a1, $-216 + 3^7\mathbb{Z}_3$	2	4	6	1512, 1512
864b1, $-13824 + 3^6\mathbb{Z}_3$	4	4	8	1512, 1512
96a1, $t \in 3\mathbb{Z}_3$	4	0	4	72, 72

TABLE S36. The division steps of Propositions 17.7 and 17.9: the dimensions of the restriction of the Selmer group to the primes above 3, of Θ and of their sum (direct in all three cases), and the numbers of cases of the enumeration and of excluded cases (none is undetermined). For 864a1 and 864b1 the elements allowed at level 0 are $c'\delta(Q_0)$ with $c' \in \{0, 1\}$. For 96a1 the levels of the tangent vectors γ_1 of the centres are: $\mathfrak{p}_1$, centre 1: level 0; $\mathfrak{p}_1$, centre 2: level 0; $\mathfrak{p}_2$, centre 1: level 18; $\mathfrak{p}_2$, centre 2: level 18; for neither centre at $\mathfrak{p}_1$ does $(\mathrm{cl}(\gamma_1), 0)$ lie in $\mathrm{loc}(\mathrm{Sel}) + \Theta$.

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